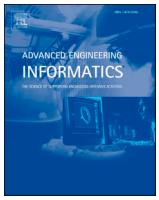
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Full length article



An interpretable physics-informed subdomain moment-enhanced adaptation network for unsupervised transfer fault diagnosis of rolling bearing

Bin Liu [a], Changfeng Yan [a,*], Chao He [b], Ming Lv [a], Jianxiong Wei [a], Lixiao Wu [a]

[a] School of Mechanical and Electronical Engineering, Lanzhou University of Technology, Lanzhou 730050, China
[b] State Key Laboratory of Advanced Rail Autonomous Operation, Beijing Jiaotong University, Beijing 100044, China

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ABSTRACT

Efficient and accurate diagnosis of rolling bearing fault is essential to prevent equipment damage. However, Intelligent Fault Diagnosis (IFD) methods are mostly data-dependent and lack interpretability, especially in the face of complex working conditions and multi-fault coupling environment, it is difficult to deal with the feature discrepancy between different distribution domains. To solve these issues, an interpretable Physics-Informed Subdomain Moment-Enhanced Adaptation Network (PISMEAN) is proposed, which aims to alleviate subdomain shift and enhance the interpretability of diagnosis process through physical convolution kernel mapping and high-order moment discrepancy metric. Firstly, the wavelet basis function is utilized for time–frequency analysis of local signal, and multiple of adaptive bandpass filtering structures are designed based on the frequency response set, thus the Physics-Informed Dynamic Convolution (PIDConv) layer is constructed, which effectively improves the correlation between the feature mapping and physical prior information. In addition, a new Local Multi-Order Moment discrepancy metric (LMOMD) is proposed, which overcomes the limitations of traditional first-order moment in low-dimensional space and ignores local nonlinear differences, bridges the center distance of sub-clusters, and further enhances the accuracy of cross-domain distribution discrepancy calculation. The fault diagnosis test is carried out on Paderborn University (PU) and Mechanical Fault Simulation (MFS) datasets, and the results show that PISMEAN has good transfer generalization performance under varying working or mixed faults conditions.

1. Introduction

As a key component of rotating machinery, the running state of rolling bearing directly affects the reliability and efficiency of the whole system. Once the bearing fails, it will not only lead to equipment downtime to increase maintenance costs, but also cause casualties even serious safety accidents [1]. However, due to the gradual development of rotating equipment to the direction of intelligence and precision, the operating environment of rolling bearings is more and more complex, for example, different rotating speeds, loads and lubrication, etc., which makes the collected vibration signals often have strong non-stationarity and high noise interference [2]. Classical signal processing and simple machine learning seem to be more difficult to adapt to the changing actual industrial scene [3–5].

With the breakthrough of Deep Learning (DL) in the fields of image recognition and Natural Language Processing (NLP) [6,7], more and more researchers have introduced classic depth models such as Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) into the field of bearing fault diagnosis. With the powerful fault feature extraction ability of Intelligent Fault Diagnosis (IFD), the complicated hyperparameter setting and prior knowledge are reduced in further, to obtain more objective and accurate diagnosis results [8–10]. Unfortunately, this kind of IFD method based on DL is a typical “black-box”, and its internal feature extraction process lacks interpretability. At the same time, there are still many shortcomings in the generalization performance of this supervised training model in the face of cross-domain data (both source and target domains are inconsistently distributed), which often leads to a sharp decline in accuracy [11]. In addition, the traditional DL method only adopts data-driven to optimize the model training process [12], and its feature mapping is difficult to capture the critical

* Corresponding author at: School of Mechanical and Electronical Engineering, Lanzhou University of Technology, Lanzhou 730050, China.
E-mail addresses: b_liu163@163.com (B. Liu), changf_yan@163.com (C. Yan), chaohe@bjtu.edu.cn (C. He).

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fault feature containing physical information, which is not conducive to achieving reliable results of fault diagnosis in the actual industrial environment.

With the wide application of Explainable Artificial Intelligence (XAI), physics-informed DL has emerged as a viable method for improving the interpretability of fault diagnosis [13]. By improving the pre-training structure or embedding the signal knowledge representing physical characteristics into the network as a regular term, the deficiency of relying on data-driven can be overcome [14]. On the other hand, by explicitly or implicitly incorporating bearing dynamics at the model level, it can help the model to Concentrate on feature that is close to the fault mechanism, and also make the learned features more explainable [15]. However, such methods not only make the network difficult to converge because of the high complexity of the training process, but also need to weigh the balance between the network and the physical regularization term, which greatly affects its diagnostic performance in the face of multi-domain distribution or complex working conditions.

Fortunately, as a type of Transfer Learning (TL), Domain Adaptation (DA) based on feature learning exhibits great potential in dealing with cross-domain distribution, and consequently, the DA based on distribution distance and adversarial strategy is proposed for fault diagnosis of rolling bearing [16]. Commonly used distribution metrics include Maximum Mean Discrepancy (MMD), Local MMD (LMMD) [17,18], aiming at overcoming the assumption that the traditional IFD methods are independent and identically distributed by minimizing the distribution discrepancy between the source and target domain, and promoting the model to maintain high fault identification performance in the target domain. However, for vibration signal with strong non-stationary characteristics [19], it is often difficult to capture more complex high-order differences between the source and target domain by adopting traditional low-order mean statistic for alignment [20]. Meanwhile, most DA pays too much attention to the discrepancy calculation, and does not make full use of the physical knowledge, which weakens the adaptability to different domain distributions. It is thus clear that despite the previous research has achieved encouraging results, there are still the following main challenges to further develop the interpretable fault diagnosis under multi-distributed data:

(1) Data-driven DA tends to ignore the physical information of faulty bearing, resulting in the inability to effectively associate key fault feature in the intermediate feature mapping stage, which makes it difficult to construct a transfer network with physical prior knowledge.

(2) There are significant differences in the data under the industrial cross-domain scene resulting in obvious distribution shift of the source and target domain, which puts forward higher requirements for evaluating the multi-domain spatial distribution metric.

(3) The complex network structure of the fault diagnosis model reduces the interpretability, which limits the reliability of the results of IFD method in practical industrial application.

Therefore, this paper proposes a new Physics-Informed Subdomain Moment-Enhanced Adaptation Network (PISMEAN) and applies it to the fault diagnosis of rolling bearing. Specifically, PISMEAN improves the performance of network cross-domain diagnosis from two dimensions: prior knowledge-enabled convolution kernel mapping and multi-order moment metric of spatial distribution discrepancy of subclasses. Firstly, physical prior convolution (PIDConv) is introduced into the first layer of the network, and the convolution kernel is initialized by adaptive bandpass filtering according to the bearing fault mechanism, which enhances the sensitivity of the model to physical information, so that the network has the ability to capture key fault feature in the early stage. Secondly, a new Local Multi-Order Moment Discrepancy (LMOMD) is proposed by using the alignment strategy of multi-order statistics. The data distribution of different domains is finely matched in high-order space, which expands the decision boundary and improves the generalization ability of Subdomain Adaptation (SA).

The following are the primary contributions to this work:

(1) A physical knowledge-embedded PIDConv is constructed. The wavelet basis function is used to analyze multi-local signal in time–frequency domain, and dynamic response frequency set is established. Based on these crucial frequencies, an adaptive bandpass filtering structure with multiple Sinc functions is designed and the convolution mapping weight is assigned. The correlation between feature mapping and physical prior information is effectively improved by PIDConv, and the interpretability of PISMEAN is increased.

(2) A new distribution metric called LMOMD is proposed. This method can measure the distribution discrepancy between the source and target domain in high-dimensional feature space, which realizes fine-grained multi-order statistical feature alignment. Compared with the traditional first-order maximum mean, LMOMD not only promotes the subdomain adaptability of the network to complex data structures, but also improves the cross-domain diagnosis effect.

(3) The transfer diagnosis performance of PISMEAN under different working conditions is evaluated, and the model's interpretability is thoroughly investigated. The results reveal that PISMEAN outperforms other IFD methods under multi-speed and mixed fault conditions.

The rest of this paper is organized as follows. Section 2 briefly reviews the research on IFD method for rolling bearing based on DA and physical information. Section 3 describes the proposed PISMEAN in detail. Section 4 presents the results of testing and discussion under different datasets. Finally, section 5 summarizes this work.

2. Related works

2.1. Fault diagnosis of rolling bearing based on unsupervised domain adaptation

The working condition of rotating machinery in the actual industrial scene is complex, which leads to that the source and target domain does not follow the same distribution, and this difference becomes more and more obvious with the change of speed or load. In order to solve this problem, IFD methods based on DA have sprung up. Unlike supervised methods that require stringent labeling of data, these methods can greatly alleviate the tedious data labeling process [21].

As mentioned in Section 1, DA is mainly divided into distance-based and adversarial-based. The former transforms different domain features into mapping space and measures the discrepancy distance with various distances. For example, Qin et al. [22] proposed an adaptive middle class distribution alignment method, which realized domain confusion through dynamic middle class and introduced AdaSoftmax at the terminal to improve the separability of the model. An et al. [11] extracted periodic and local features through gaussian mixture transform layer, and used domain alignment term to map features to shared space. Cui et al. [23] proposed a graph DA method to improve the ability to deal with non-Euclidean data, and enhance the discrepancy calculation through graph Laplacian regularization MMD. Yang et al. [24] extracted cross-domain data features by constructing a shared residual network, and collected target domain samples by using a clustering module, so as to realize the correction of joint distribution offset by associating source domain labels. Qian et al. [25] proposed a Relational Transfer Domain Generalized Network (RTDGN), which used two-stage DA to enhance the domain identification ability of the network, and designed the inverse entropy loss to improve the accuracy of the classifier. Yu et al. [26] realized local–global feature extraction by combining time–frequency features and global statistical matrix, and adopted Multi-Kernel MMD (MK-MMD) to improve the learning ability of the network.

The adversarial-based DA learns from the Generative Adversarial Network (GAN) idea, and plays a game between feature extractor and domain discriminator, thus reducing the negative impact brought by

domain shift. For example, Gao et al. [18] utilized adversarial TL to obtain generalized feature, and proposed a joint distributed DA to further promote the learning of domain invariant feature. Chen et al. [27] used edge adversarial module to calculate the adversarial loss, and constructed an internal adversarial module to perform feature extraction, which enhanced the confusion performance of network. Li et al. [28] proposed a data-combined labeled domain expansion method, and improved the feature extraction ability of the model. Deng et al. [29] used progressive learning strategy to disperse unknown samples to improve the separability between classes. Fan et al. [17] proposed a deep mixed domain generalization network, which applied data expansion to feature space and introduced the adversarial perturbation for the computation of different distributions. Wang et al. [30] used the simulation data to construct model, and combined domain adversarial strategy with WD to propose a new TL network.

Meanwhile, in order to learn the local feature discrepancy in more detail, SA has been gradually focused on in recent years. As shown in Fig. 1, global DA aligns the overall feature distributions of the source and target domains, regardless of the differences in category labels or substructures within the data, whereas SA aligns the features of each category separately to ensure that the feature distributions of the corresponding categories in the source and target domains can match each other. By comparison, global DA only pays attention to the consistency of the overall distribution, while SA can capture the fine-grained differences of each category to avoid confusion between features of different categories during the alignment process. For instances, Xiao et al. [31] used dilated convolution to extract multi-scale features, and allocated channel weights through classical Attention Mechanism (AM), which provided critical discrepancy information for different subdomains. Liu et al. [32] proposed deep SA mechanism, which combined statistical indicators and geometric features to construct a subdomain shift metric module. Shi et al. [33] introduced GAN into SA to extract deep feature, and calculated subdomain discrepancy through LMMD. Zhuang et al. [34] expressed the subdomain feature via manifold learning, and embedded the graph structure into the subdomain network to predict the health state. Luo et al. [35] proposed an order tracking method of domain transformation, and embedded the constructed harmonic extraction method into the network. Yuan et al. [36] introduced clustering local model based on active learning to enrich training data, and used standardized LMMD to calculate subdomain discrepancy.

Although the above methods further expand the problem of fault diagnosis under complex data and achieve excellent results, it is undeniable that the use of complex network structure in many studies greatly weakens the transparency of the diagnosis process, resulting in poor interpretability of the methods. Besides, single and low-order metrics seem to be difficult to meet the needs of calculating distribution distance when facing high-dimensional complex data, which limits the transfer performance of the model.

2.2. Fault diagnosis of rolling bearing based on physical knowledge

Because of internal defects, the vibration signal will exhibit periodic impact. This typical impulse response frequently includes physical feature that can characterize the fault type. Obviously, learning and extracting such information accurately can not only improve the diagnostic effect of IFD, but also provide better interpretability to the model. Naturally, physical-informed network is introduced into the field of fault diagnosis as a primary research focus, and the topic can be divided into post-hoc and ante-hoc interpretable models. The former hardly changes the network structure, but deduces the diagnosis process through the output results, including Class Activation Mapping (CAM) and Shapley additional explanations (SHAP) [37].

For illustration, Kim et al. [38] identified fault types based on frequency characteristics, and visualized the decision criteria of the model in frequency domain. Chen et al. [39] used the multi-branch AM to focus on the key feature of vibration signal, and analyzed the diagnosis results with GS-CAM. Yang et al. [40] used Short-Time Fourier Transform (STFT) to identify vibration signal, and visually explained the learned feature. Gwak et al. [41] proposed a decision boundary analysis model, and conducted power interference centering on the main category frequency to obtain training data. Compared with post-result analysis, ante-hoc interpretable model tends to understand the diagnosis process from the essential inherent attributes, and has more complicated knowledge embedding behavior. Liu et al. [42] proposed a subdomain-enhanced adaptive network, which enhanced physical knowledge through sparse subsegment noise reduction layer, and calculated the discrepancy of different subdomains by using improved LMMD, thus promoting the transfer diagnosis performance of the network. He et al. [43]combined the differentiable STFT and physical prior frequency to measure the loss, so as to learn the domain invariant feature under time-varying speed. Li et al. [44] utilized the wavelet packet convolutional layer to constrain different mapping kernels, and reduced the noise component in feature mapping by soft threshold. Liu et al. [45] proposed a dynamic recalibration mechanism to suppress noise interference, and learned different levels of fault information through a new feature extraction module. Li et al. [46] generated signal nodes through learnable wavelet operators, then combined statistical feature and classifier to make the network interpretable.

Undoubtedly, the above research has made great contributions to improving the interpretability of IFD methods. However, most of them involve complex feature mapping, which weakens the representation of the original initial signal. In addition, there is a lack of analysis of domain-identifiable feature and network generalizability when facing complex cross-domain data.

3. Methodology

In the process of applying DL to the field of fault diagnosis, the traditional CNNs often rely on the randomly initialized convolutional kernels when facing vibration signal, and lack the active mining of the

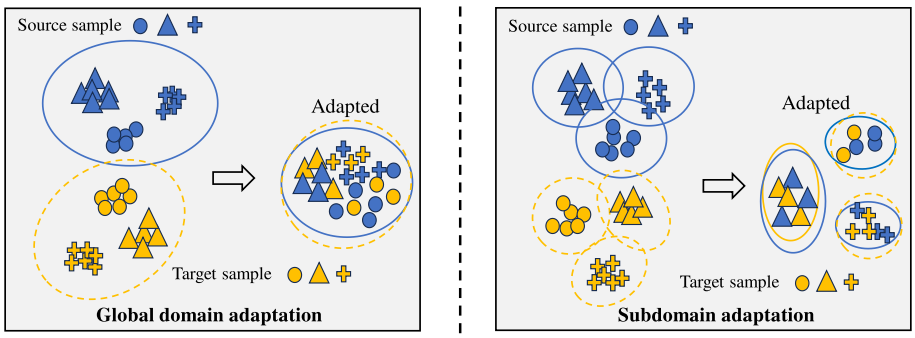


Fig. 1. Description of different DA methods.

physical characteristic of the signal, which leads to poor interpretability. Therefore, in this study, PISMEAN is proposed. By introducing the convolution layer of physical information, the kernel is dynamically initialized by using the time–frequency feature of the signal, and the perception ability of the convolution layer to key fault frequencies is enhanced. In addition, the multi-order moment-enhanced discrepancy metric is adopted, which avoids the problem of inaccurate local feature alignment caused by the traditional method that only depends on the mean value, and further improves the feature transfer ability of the network. In this section, the overall process of the proposed method, the design of PIDConv, the new multi-subdomain metric, and its application are described in detail.

3.1. Physics-informed dynamic convolution layer with weight initialization

As a vital step in fault diagnosis, accurately extracting feature with physical meaning can effectively improve the accuracy of diagnosis results. The fault characteristic frequency is determined by the contact behavior between local defects and different components. As shown in Eq. (1), the information that can reflect the fault mechanism is often used in signal processing methods to identify different types of faults. However, the data in the actual industrial scene are interfered by many factors, and retaining correct feature information often depends on a lot of prior knowledge and expert experience, which is obviously not conducive to the requirements of efficient diagnosis.

$$\begin{cases} f_o = \dfrac{n_b \times F}{2}\left(1 - \dfrac{d}{D_p}\cos\theta\right) \\ f_i = \dfrac{n_b \times F}{2}\left(1 + \dfrac{d}{D_p}\cos\theta\right) \\ f_b = \dfrac{D_p \times F}{2d}\left(1 - \left(\dfrac{d}{D_P}\right)^2 \cos^2\theta\right) \end{cases} \tag{1}$$

where, f_o, f_i, f_b represent the characteristic frequencies of Outer Race (OR), Inner Race (IR) and rolling element fault, respectively. n_b is the number of rolling elements, d is the diameter, θ is the contact angle, and D_P is the diameter of the pitch circle.

Algorithm 1 Physics-informed convolution weight initialization

Input: N_i: input channels, No: output channels, K: kernel size, n: discrete time index, f_s: sampling rate
Output: $W_{i,j}$: Convolutional weights.
1: The set of construction target frequency $\left\{f_k^i\right\}$ is calculated by Eqs. (4)–(7).
2: The dynamic filtering kernel $g_k^i(n)$ is generated by substituting $\left\{f_k^i\right\}$ into the center frequency fc of different Sinc kernels via Eq. (8).
3: Assigning $g_k^i(n)$ to the convolutional kernel weights completes the weight initialization via Eq. (10).
4: The PIDConv is continuously fine-tuned during training by BP, $W_{ij}(n+1) = W_{ij}(n) - \gamma\dfrac{\partial L}{\partial W_{ij}}$, where γ is the learning rate and L is the network loss.

With its powerful feature extraction capability, DL-based IFD be trained to extract useful feature from the original signal. However, most CNNs are only driven by data, and this randomly initialized convolutional kernel does not actively pay attention to the fault characteristic frequency in the vibration signal. Meanwhile, inappropriate network weight will also lead to the model falling into the local optimal solution during Back Propagation (BP), which makes the diagnosis effect unsatisfactory. On the other hand, the multi-layer network mapping will weaken the expression of initial information. Although the AM can adjust the weight distribution, this auxiliary structure based on feature mapping cannot directly mine and use the physical characteristic of signal [45].

Therefore, to improve the interpretability and feature extraction performance, it is important to analyze the frequency domain of the signal to embed the physical prior knowledge, i.e., to empower the prior knowledge of signal to the convolution map in the form of adaptive filtering. Inspired by this, as shown in Fig. 2, the PIDConv layer with weight initialization is proposed. Distinct from the previous approaches which rely on fixed convolution kernel to map different information, PIDConv doubly resolves the vibration signal in time–frequency domain, establishes a physical prior bandpass filtering structure, and inferred the optimal network weight by adaptively adjusting parameters through BP. It not only overcomes the shortcomings that the initialization methods such as random and Xavier initialization [47] would fail to consider the physical feature of bearing fault, but also solves the tedious hyperparameter setting in traditional signal processing methods. This prior knowledge based on signal processing is used to initialize the weights of PIDConv, to reflect the time–frequency domain changes of different vibration signal more accurately. At the same time, with the continuous training process, convolution weights can be further fine-tuned according to gradient to meet the requirements of different feature drift or complex scenarios.

Specifically, the signal collected by acceleration sensor is $X = [x^1, x^2, \cdots, x^n]$, where $x^i \in \mathbb{R}^{1\times l\times 1}(i = 1, 2, 3\cdots n)$ represents the i-th sample. The sample data can be modeled as the superposition of target characteristic frequency and noise.

$$\begin{cases} x^i(t) = s^i(t) + o^i(t) \\ s^i(t) = \sum_{j=1}^{m} A_j \sin\left(2\pi f_j t + \varphi_j\right) \end{cases} \tag{2}$$

where, $s^i(t)$ represents the target signal, $o^i(t)$ is the random noise, f_j, A_j, φ_j denote the frequency, amplitude and phase associated with the bearing fault, respectively. Firstly, to analyze the frequency characteristics of fault signals in different time scales. The real Laplace wavelet basis function is used to respond locally to different time–frequency to extract the multi-frequency components. The time domain expression of wavelet kernel as shown in Eq. (3) [48].

$$\psi(t) = Ae^{\frac{-\xi}{\sqrt{1-\xi^2}}\times 2\pi f(t-\tau)}\sin[2\pi f(t-\tau)] \tag{3}$$

where, A is the wavelet normalization factor, f represents the sampling frequency, ξ denotes the damping ratio, and τ is the time parameter.

The vibration signal contains the superposition of multiple frequency components, in order to calculate the frequency response under different scales, it is necessary to determine the main frequency components by time–frequency localization analysis through the wavelet kernel, so as to realize the perception of the different frequency characteristics containing fault information under the frequency domain criterion. The convolution process of sample signal x^i with the wavelet basis function as shown in Eq. (4).

$$C^i(t,f) = x^i(t) * \psi(t) = \int_{-\infty}^{\infty} x^i(\tau)\cdot\psi(t-\tau)d\tau \tag{4}$$

where, $C^i(t,f)$ denotes the wavelet coefficients, * is the convolution operation. Similarly, in practice, discrete signal $x^i(n)$ is usually used to calculate wavelet coefficients.

$$C^i(n,f) = \sum_{k=0}^{l-1} x^i(k)\cdot\psi(n-k) = \sum_{k=0}^{l-1} x^i(k)\cdot\left[Ae^{\frac{-\xi}{\sqrt{1-\xi^2}}\times 2\pi f|n-k|}\sin(2\pi f(n-k))\right] \tag{5}$$

Obviously, $C^i(n,f_s)$ can clearly characterize the amplitude and energy distribution of signals at different sampling frequencies f_s. Then, the frequency spectrum of the convolution result is calculated by Fourier transform to describe the distribution state of different frequency

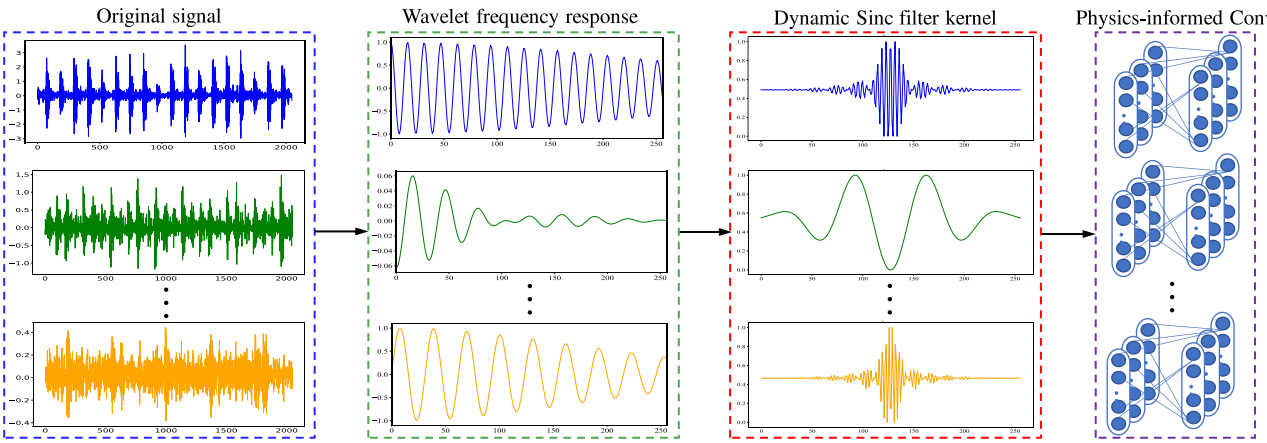


Fig. 2. PIDConv with adaptive bandpass filtering.

responses.

$$C^i(k)=\sum_{k=0}^{N-1} C^i(n,f)\cdot e^{\frac{-j2\pi kn}{N}}, k=0,1,...,N-1 \tag{6}$$

where, N is the signal length.

Then, the frequency component with high energy concentration in the amplitude spectrum $|C^i[k]|$ is selected as the main target frequency, and the dynamic frequency response set $\left\{f_k^i\right\}$ can be obtained.

$$\begin{cases} |C^i(k)| = \sqrt{\mathrm{Re}(C^i(k))^2+\mathrm{Im}(C^i(k))^2} \\ \mathrm{Re}(C^i(k)) = \sum_{n=0}^{N-1} C^i(n)\cdot\cos\left(\frac{2\pi kn}{N}\right) \\ \mathrm{Im}(C^i(k)) = -\sum_{n=0}^{N-1} C^i(n)\cdot\sin\left(\frac{2\pi kn}{N}\right) \end{cases} \tag{7}$$

where, the real and imaginary parts correspond to the cosine and sine components of the frequency components, respectively.

As a key parameter to construct bandpass filtering, the central frequency f_c is often set as the fault characteristic frequency [49]. However, as shown in Eq. (1), this parameter setting method will weaken the diagnosis effect in cross-domain data with obvious speed difference. Therefore, PIDConv skillfully applies the dynamic frequency set for the Sinc kernel f_c respectively, realizes the purpose of multi-channel filtering in the form of convolution kernel, and extracts the physical information of different scales through the network. This dynamic filtering strategy allows PIDConv to automatically adapt to variations in rotational speed, as the extracted frequency response set reflects the actual operating condition of each sample. In discrete data, Sinc kernel g(n) of bandpass filter can be defined as Eq. (8).

$$g(n)=\frac{\sin\left(\frac{2\pi f_\theta n}{f_s}\right)}{\pi n}\cdot\cos\left(\frac{2\pi f_c n}{f_s}\right)\cdot\omega(n) \tag{8}$$

where, f_θ is the bandwidth, $\omega(n)$ is the smoothing window function. Introducing $\omega(n)$ can avoid spectrum leakage when the Sinc kernel is truncated. Common window functions include Hamming window ω_H and Gaussian window ω_G, as shown in Eq. (9).

$$\begin{cases} \omega_H = 0.54-0.46\cos\left(\frac{2\pi n}{N-1}\right) \\ \omega_G = \exp\left(\frac{-n^2}{2\sigma^2}\right), \sigma=\frac{N}{6} \end{cases} \tag{9}$$

where, n represents the length of the Sinc kernel. Then, $\left\{f_k^i\right\}$ is introduced into Eq. (8), the corresponding dynamic filtering kernel $g_k^i(n)$ will be generated for each target frequency f_k^i and used for the initialization of convolution kernel. Accordingly, the convolution output with physical information can be expressed as shown in Eq. (10).

$$\begin{cases} W_{i,j} = g_k^i(n) \\ g_k^i(n) = \frac{\sin\left(\frac{2\pi f_\theta n}{f_s}\right)}{\pi n}\cdot\cos\left(\frac{2\pi f_k^i n}{f_s}\right)\cdot\omega(n) \\ y_i(n) = \sum_{j=1}^{C}\sum_{m=0}^{N-1} W_{i,j}\cdot x_i(n)+b_i \end{cases} \tag{10}$$

where, y_i denotes the i-th channel convolution output, $W_{i,j}$ is the weights, b_i represents the bias, and C is the number of channels.

The proposed PIDConv addresses the limitations of traditional data-driven approaches by embedding physical prior knowledge into the convolutional process. Specifically, the bandpass filtering structure, constructed through the combination of wavelet-based frequency response analysis and dynamic Sinc kernel initialization, enables effective utilization of time–frequency domain characteristics. This design guides the convolutional kernels to focus on fault-relevant spectral features, thereby improving the interpretability and relevance of the extracted representations. Furthermore, the use of multiple parallel filtering kernels expands the receptive field of the network and mitigates the constraints of single-band filtering commonly found in conventional signal processing methods. Importantly, the initialized filters are trainable and continuously refined through backpropagation, which enhances the adaptability of the model to varying working conditions and improves cross-domain diagnostic performance. The detailed operational steps of the PIDConv layer are summarized in Algorithm 1.

3.2. Local multi-order moment discrepancy

Bearing vibration signals, as a typical type of non-stationary data, exhibit clear nonlinear characteristics, which lead to more complex data distributions in high-dimensional space. LMMD, which only relies on first-order statistics, does not account for changes in the high-order statistical characteristics of each local domain, thereby limiting the network's ability to transfer subdomain knowledge. Based on this, this study proposes the LMOMD. By introducing high-order moments (second-order variance), the kernel function can calculate more comprehensive subdomain distributions, especially in high-dimensional space, which can effectively avoid the subtle distribution discrepancy that cannot be reflected by low-order statistics. In the target domain, since

the true labels are not available, the pseudo-labels predicted by the classifier are usually utilized to define the subdomain category c. Specifically, each target sample is assigned to the c-th subdomain based on its predicted category, i.e., $y_i^T = \text{argmax} f(x_i^T)$. Therefore, when calculating LMOMD, pseudo-labels provide the necessary guidance for cross-domain alignment of the corresponding subcategory distributions.

Specifically, assuming that the distribution of the source and target domain is P_S and P_T, respectively.

$$\begin{cases} D_S = \{x_i^S\}_{i=1}^m, x_i^S \sim P_S \\ D_T = \{x_i^T\}_{j=1}^n, x_i^T \sim P_T \end{cases} \tag{11}$$

where, x_i^S and x_i^T represent the source and target domain samples, respectively. The core idea of MMD is to map the data to the Reproducing Kernel Hilbert Space (RKHS), so as to calculate the difference of kernel mean embedding in different domains, i.e., the kernel function $k(x,y) = \langle \varnothing(x), \varnothing(y) \rangle_{\mathscr{H}}$, where $\varnothing(x) \in \mathscr{H}$ represents the corresponding feature mapping [42]. Naturally, the source domain mean value μ_S and target domain mean value μ_T in RKHS are defined as Eq. (12).

Algorithm 2 The training process of PISMEAN

Input: D_S: source domain; D_T: target domain; γ: learning rate of θ; batch size; M: model; λ: the trade-off coefficient.
Output: Parameters of PISMEAN: f.

```
for M not converged do
  for epoch = 0 to eraly_stopping do
    B batch samples (x_i^s, y_i^s) from D_S
    B batch samples x_i^t from D_T
    Extract fault features of subdomains using M
    Output pseudo labels of x_i^t: y_i^t
    Calculate CE loss l_c and LMOMD loss l_LMOMD by Eq. (25)
    Calculate and update λ by Eq. (26)
    Optimize the parameters of PISMEAN: f by Eq. (27)
  end for
end for
```

$$\begin{cases} \mu_S = \mathbb{E}_{x\sim P_S}[\phi(x)] \\ \mu_T = \mathbb{E}_{x\sim P_T}[\phi(x)] \end{cases} \tag{12}$$

The unbiased estimation MMD is defined as Eq. (13) [42].

$$\begin{aligned} \text{MMD}^2(D_S, D_T) &= \left\| \frac{1}{n_s}\sum_{i=1}^{n_s}\phi(x_i) - \frac{1}{n_t}\sum_{j=1}^{n_t}\phi(x_j) \right\|_{\mathscr{H}}^2 \\ &= \frac{1}{n_s^2}\sum_{i=1}^{n_s}\sum_{j=1}^{n_s}k\left(x_i^s, x_j^s\right) + \frac{1}{n_t^2}\sum_{i=1}^{n_t}\sum_{j=1}^{n_t}k\left(x_i^t, x_j^t\right) - \frac{2}{n_s n_t}\sum_{i=1}^{n_s}\sum_{j=1}^{n_t}k\left(x_i^s, x_j^t\right) \end{aligned} \tag{13}$$

where, n_s, n_t denote the number of samples, respectively. Meanwhile, LMMD introduces the concept of subdomain on the basis of the above calculation method, and considers the local multi-category discrepancy more finely. Similarly, the mean value of the c-th subdomain in RKHS is expressed as Eq. (14).

$$\begin{cases} \mu_S^c = \dfrac{1}{m_c}\displaystyle\sum_{i=1}^{m_c}\phi\left(x_i^{S,c}\right) \\ \mu_T^c = \dfrac{1}{n_c}\displaystyle\sum_{j=1}^{n_c}\phi\left(x_j^{T,c}\right) \end{cases} \tag{14}$$

Therefore, LMMD is to add the mean values of each subdomain to measure the local distribution discrepancy.

$$\text{LMMD}(D_S, D_T) = \sum_{c=1}^{C}\text{MMD}(D_S, D_T) \tag{15}$$

Obviously, the above traditional discrepancy measures only consider the first-order statistics. The proposed LMOMD retains the domain confusion ability of low-dimensional space, and further considers the second-order moment variance discrepancy. In particular, the proposed method also follows the kernel function mapping, thus expanding the original LMMD's ability to discriminate high-dimensional spatial distribution. Similarly, the variance of source and target domain in the c-th subdomain is defined as Eq. (16).

$$\begin{cases} \sigma_S^c = \mathbb{E}_{x\sim P_S^c}\left[\left\|\phi(x) - \mu_S^c\right\|_{\mathscr{H}}^2\right] \\ \sigma_T^c = \mathbb{E}_{x\sim P_T^c}\left[\left\|\phi(x) - \mu_T^c\right\|_{\mathscr{H}}^2\right] \end{cases} \tag{16}$$

The norm definition of $\phi(x)$ in the kernel space as shown in Eq. (17).

$$\|\phi(x)\|_{\mathscr{H}}^2 = \langle \phi(x), \phi(x) \rangle_{\mathscr{H}} = k(x,x) \tag{17}$$

Accordingly, the expansion of σ_S^c can be expressed as shown in Eq. (18).

$$\begin{aligned} \sigma_S^c &= \mathbb{E}_{x_S^c\sim P_S^c}\left[\langle \phi(x_S^c) - \mu_S^c, \phi(x_S^c) - \mu_S^c \rangle_{\mathscr{H}}\right] \\ &= \mathbb{E}_{x_S^c\sim P_S^c}\left[\langle \phi(x_S^c), \phi(x_S^c) \rangle_{\mathscr{H}} - 2\langle \phi(x_S^c), \mu_S^c \rangle_{\mathscr{H}} + \langle \mu_S^c, \mu_S^c \rangle_{\mathscr{H}}\right] \\ &= \mathbb{E}_{x_S^c\sim P_S^c}\left[\left\|\phi(x_S^c)\right\|_{\mathscr{H}}^2 - 2\langle \phi(x_S^c), \mu_S^c \rangle_{\mathscr{H}} + \left\|\mu_S^c\right\|_{\mathscr{H}}^2\right] \end{aligned} \tag{18}$$

The definition of μ_S^c is given in Eq. (19).

$$\mu_S^c = \mathbb{E}_{x_S^c\sim P_S^c}\left[\phi(x_S^c)\right] \tag{19}$$

Consequently, σ_S^c can be ultimately defined as shown in Eq. (20).

$$\sigma_S^c = \mathbb{E}_{x_S^c\sim P_S^c}\left[\left\|\phi(x_S^c)\right\|_{\mathscr{H}}^2\right] - \left\|\mu_S^c\right\|_{\mathscr{H}}^2 \tag{20}$$

Similarly, the definition of σ_T^c can also be expressed as shown in Eq. (21).

$$\sigma_T^c = \mathbb{E}_{x_T^c\sim P_T^c}\left[\left\|\phi(x_T^c)\right\|_{\mathscr{H}}^2\right] - \left\|\mu_T^c\right\|_{\mathscr{H}}^2 \tag{21}$$

In practice, the expectation is estimated from finite samples, as illustrated in Eq. (22).

$$\begin{cases} \mathbb{E}_{x_S^c\sim P_S^c}\left[\left\|\phi(x_S^c)\right\|_{\mathscr{H}}^2\right] = \dfrac{1}{m_c}\displaystyle\sum_{i=1}^{m_c}\left\|\phi(x_i^{S,c})\right\|_{\mathscr{H}}^2 = \frac{1}{m_c}\sum_{i=1}^{m_c}k(x_i^{S,c}, x_i^{S,c}) \\ \mathbb{E}_{x_T^c\sim P_T^c}\left[\left\|\phi(x_T^c)\right\|_{\mathscr{H}}^2\right] = \dfrac{1}{m_c}\displaystyle\sum_{i=1}^{m_c}\left\|\phi(x_i^{T,c})\right\|_{\mathscr{H}}^2 = \frac{1}{m_c}\sum_{i=1}^{m_c}k(x_i^{T,c}, x_i^{T,c}) \end{cases} \tag{22}$$

Therefore, the subdomain discrepancy is estimated by the kernel function as follows:

$$\begin{cases} \sigma_S^c = \dfrac{1}{m_c}\displaystyle\sum_{i=1}^{m_c}k(x_i^{S,c}, x_i^{S,c}) - \left\|\mu_S^c\right\|_{\mathscr{H}}^2 \\ \left\|\mu_S^c\right\|_{\mathscr{H}}^2 = \dfrac{1}{m_c^2}\displaystyle\sum_{i,\tilde{i}}k(x_i^{S,c}, x_{\tilde{i}}^{S,c}) \\ \sigma_T^c = \dfrac{1}{n_c}\displaystyle\sum_{j=1}^{n_c}k\left(x_j^{T,c}, x_j^{T,c}\right) - \left\|\mu_T^c\right\|_{\mathscr{H}}^2 \\ \left\|\mu_T^c\right\|_{\mathscr{H}}^2 = \dfrac{1}{n_c^2}\displaystyle\sum_{j,\tilde{j}}k\left(x_j^{T,c}, x_{\tilde{j}}^{T,c}\right) \end{cases} \tag{23}$$

The variance distribution of different domains can be calculated by substituting the kernel mean in Eq. (14) into Eq. (23). Therefore, the LMOMD based on multi-order moment discrepancy measure is defined as Eq. (24).

$$\mathrm{LMOMD}(D_S, D_T) = \left(\sqrt{ \frac{1}{m_c^2} \sum_{c=1}^{C} \sum_{i,\tilde{i}} k\left(x_i^{S,c}, x_{\tilde{i}}^{S,c}\right) + \frac{1}{n_c^2} \sum_{c=1}^{C} \sum_{j,\tilde{j}} k\left(x_j^{T,c}, x_{\tilde{j}}^{T,c}\right) - \frac{1}{n_c m_c} \sum_{c=1}^{C} \sum_{i,j} k\left(x_i^{S,c}, x_j^{T,c}\right) } \right) + \left| \left(\frac{1}{m_c} \sum_{c=1}^{C} \sum_{i=1}^{m_c} k\left(x_i^{S,c}, x_i^{S,c}\right) - \frac{1}{m_c^2} \sum_{c=1}^{C} \sum_{i,\tilde{i}} k\left(x_i^{S,c}, x_{\tilde{i}}^{S,c}\right) \right) - \left(\frac{1}{n_c} \sum_{c=1}^{C} \sum_{j=1}^{n_c} k\left(x_j^{T,c}, x_j^{T,c}\right) - \frac{1}{n_c^2} \sum_{c=1}^{C} \sum_{j,\tilde{j}} k\left(x_j^{T,c}, x_{\tilde{j}}^{T,c}\right) \right) \right| \tag{24}$$

From the above calculation process, LMOMD can improve the sensitivity to the non-uniform changes of subclasses or subclusters. Secondly, LMOMD is an extension of LMMD, i.e., combining first-order and second-order moments as comprehensive evaluation indicators.

As illustrated in Fig. 3, the distribution discrepancy between the source and target domains in high-dimensional feature space cannot be sufficiently characterized using only first-order metrics. In such scenarios, relying solely on mean alignment may lead to misaligned subdomains, especially when class-wise variances differ significantly. By incorporating second-order statistics, the proposed LMOMD metric not only captures the central tendency of each subdomain, but also models the relative dispersion and structural relationships between samples in the mapped space. This provides a more comprehensive and discriminative criterion for domain alignment, particularly effective for complex data that deviate from simple Gaussian assumptions. Moreover, the use of higher-order moments offers a promising strategy for enhancing adaptability under varying and challenging working conditions.

3.3. Architecture and training of PISMEAN

The model structure of PISMEAN is shown in Fig. 4, including adaptive filtering PIDConv layer, feature extraction module and LMOMD with multi-local subdomain discrepancy distribution alignment. As shown in Fig. 4, LMOMD is employed to perform subdomain adaptation between the source and target domain features, enabling fine-grained subdomain feature alignment. The required pseudo-labels for the target domain are generated by the model's classifier. The specific network parameters are shown in Table 1. Firstly, PIDConv as the first layer of the network by filtering the original signal processing, so that the convolution kernel can directly extract the physical information with key features instead of simply randomly initializing, overcoming the traditional CNN first layer of the convolution relies on data-driven approach to training, thus reducing noise and improving the interpretability of the diagnosis process. Secondly, embedding the network in the form of prior knowledge can fully adapt to the subsequent depth feature extraction module, which is helpful for more classic backbone applications, such as WDCNN, ResNet, MCNN. Finally, to calculate the distribution discrepancy of multi-class subdomains more comprehensively, the second-order moment is introduced to further calculate the internal structural relationship of different samples in high-dimensional space, i.e., the mean and variance of data are jointly calculated by LMOMD, so as to align the distribution of different domains. The main goal of PISMEAN is to optimize the source domain loss l_c and the LMOMD loss l_{LMOMD} between different subdomains. Therefore, the overall loss of the network can be expressed as Eq. (25).

$$l = l_c + \lambda l_{LMOMD} \tag{25}$$

where, λ is a trade-off coefficient.

$$\lambda = \frac{-4}{\sqrt{\frac{epoch}{\max - epoch} + 1}} + 4 \tag{26}$$

The training parameter f corresponding to the gradient change of the network can be expressed by Eq. (27).

$$\nabla l = \nabla l_c + \lambda \nabla l_{LMOMD} \tag{27}$$

The training parameter f can be controlled by the optimizer Adam.

$$f \leftarrow f - \left[\gamma \left(\frac{\partial l}{\partial \theta} + \lambda (\nabla l_{LMOMD}) \right) \right] \tag{28}$$

where, γ represent the learning rate.

3.4. Transfer diagnosis process based on PISMEAN

The fault diagnosis process as shown in Fig. 5, which not only includes bearing fault scenarios with different loads and rotating speeds, but also simulates mixed fault in the laboratory for transfer diagnosis that is more in line with industrial scenarios. Compared to most diagnostic tasks in the past, the inclusion of single and compound fault is more challenging for IFD.

(1) The condition monitoring terminal collects the original vibration signal of the rolling bearing through the sensor arranged in the equipment.

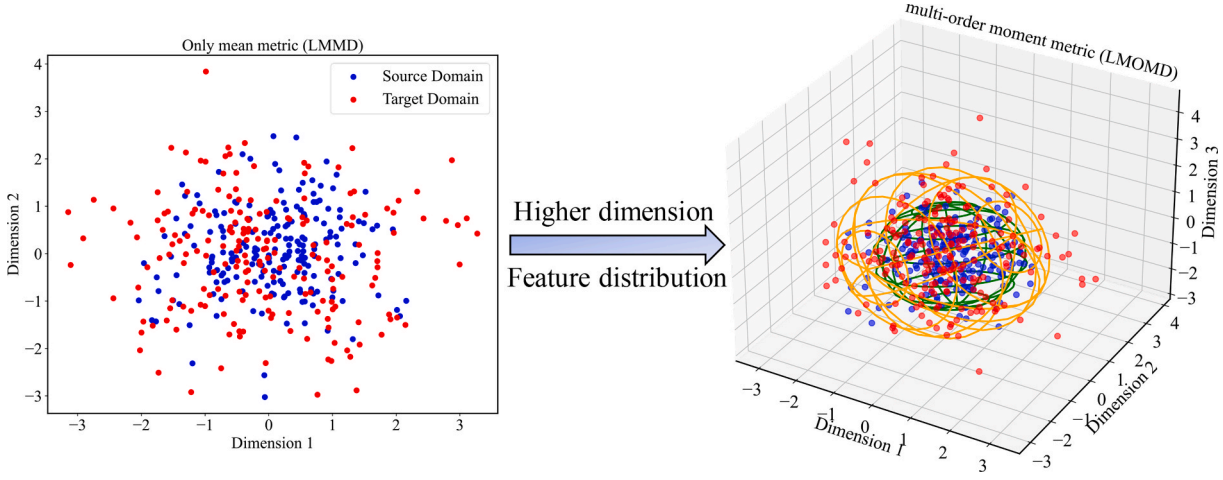


Fig. 3. High-dimensional spatial feature distribution.

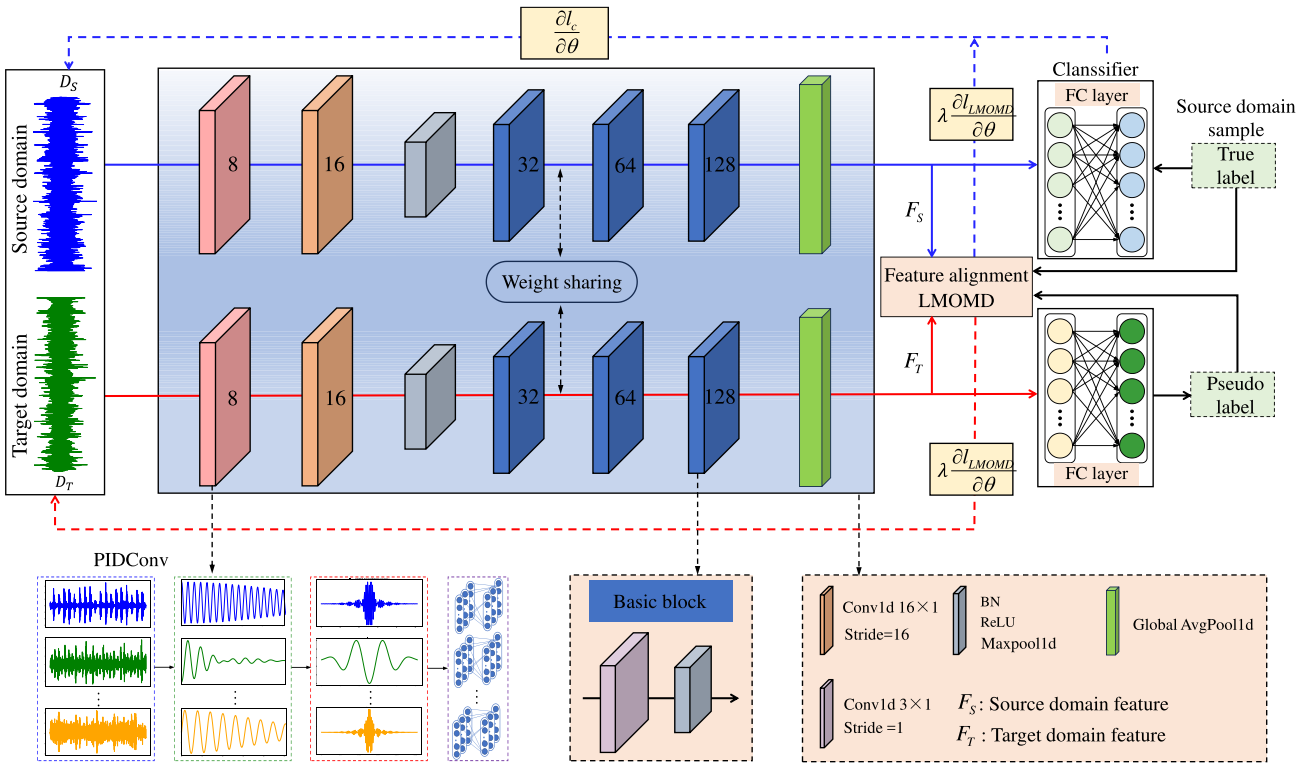


Fig. 4. PISMEAN network architecture with subdomain adaptation.

Table 1
Parameters of PISMEAN.

Type of layer	Specific operation	Channel/Kernal size/ Stride/Padding	Input size	Output size
PIDConv-1	Adaptive filter	8/256/1/Same	(1,2048,1)	(8,2048,1)
Conv-2	Convolution	16/64/16/24	(8,2048,1)	(16,128,1)
	BN	/	(16,128,1)	(16,128,1)
	Max-pooling	16/2/2	(16,128,1)	(16,64,1)
Conv-3	Convolution	32/3/1/1	(16,64,1)	(32,64,1)
	BN	/	(32,64,1)	(32,64,1)
	Max-pooling	32/2/2	(32,64,1)	(32,32,1)
Conv-4	Convolution	64/3/1/1	(32,32,1)	(64,32,1)
	BN	/	(64,32,1)	(64,32,1)
	Max-pooling	64/2/2	(64,32,1)	(64,16,1)
Conv-5	Convolution	128/3/1/1	(64,16,1)	(128,16,1)
	BN	/	(128,16,1)	(128,16,1)
	Max-pooling	128/2/2	(128,16,1)	(128,8,1)
GAP/FC (activation = 'Softmax')		Num-class/*/1	128	Num-class

(2) Divide the collected signals into different samples, including source domain data with labels and target domain data without labels.
(3) The sample data is input to PISMEAN. Firstly, as shown in Eq. (6)-(10), PIDConv at the first layer of the network performs adaptive bandpass filtering and physical information extraction. Then, the deep fault features are extracted by multi-layer convolution nonlinear mapping. Finally, as shown in Eq. (16)-(27), the distribution discrepancy of different subdomains are calculated by using the proposed LMOMD, so as to realize the DA process.
(4) As shown in Algorithm 2, the BP of the network is regulated by the total loss function, and Adam is utilized to optimize the model training.
(5) Repeat steps 3 and 4 until the training loss value meets the setting in the early stopping, and save the model parameters.
(6) Using different datasets to test the trained PISMEAN and compare it with other IFD methods.

4. Results analysis and discussions

The Mechanical Fault Simulation (MFS) datasets in our laboratory and the bearing fault public datasets by the Paderborn University (PU) are used to verify the proposed PISMEAN, in which the diagnosis scenario includes not only the cross-domain conditions of speed and load changes, but also involve mixed faults (both single and compound faults). In order to ensure a fair comparison with other IFD methods, the learning rate and batch size are uniformly set to 0.001 and 64. At the same time, in this study, the training samples are clearly expressed as N_t, which is convenient for quantitative analysis of the influence of different sample numbers on the diagnosis effect.

4.1. Data description

As shown in Fig. 6, the PU bearing test bench consists primarily of a test motor, rotating shaft, bearing and load motor. Different variables (speed, load, torque) are varied to model changes in the bearings' health status under various operating situations. The PU datasets contain different damage types and fault generation paths, which makes the data sets more challenging [50]. The sample data are collected by the acceleration sensor in the bearing cover, and the sampling rate is 64 kHz, which is divided into four working conditions according to the different operating variables. The description of PU datasets is list in Table 2.

The MFS test rig in our laboratory is shown in Fig. 7. The test bearings are all etched by laser, and the fault types are OR, IR, ball and compound faults [45]. The width and depth of local defects are 0.6, 1.2, 1.8 and 0.25 mm respectively. The sampling rate is 15.3 kHz, and the description of MFS datasets is list in Table 3.

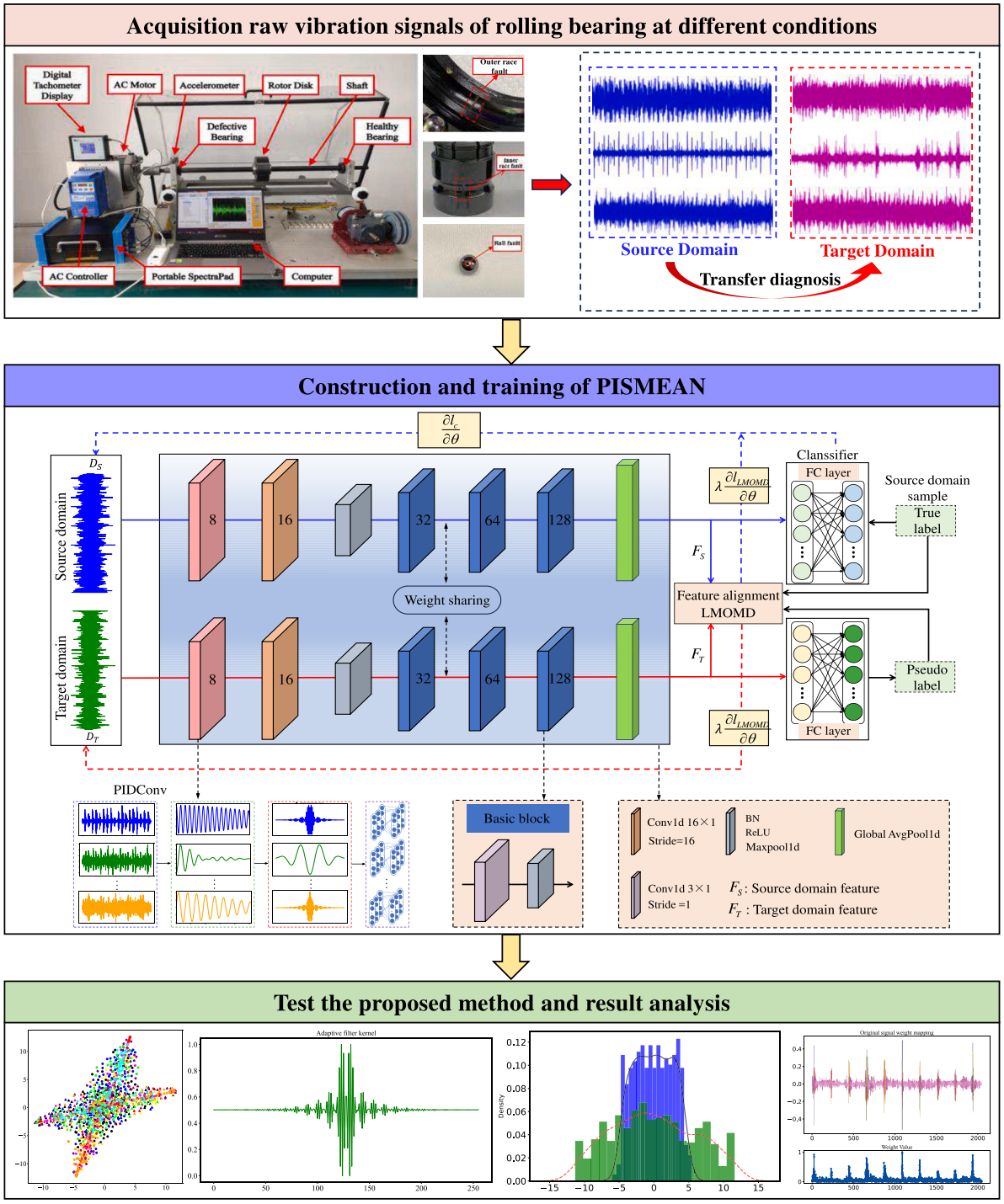


Fig. 5. Fault diagnosis process based on PISMEAN.

4.2. Comparison of existing approaches

The comparison methods include ISEANet [42] (ILMMD), EWSNet [48] (MMD), DSAN [51] (LMMD), MRANet [45] (MMD), DDC [52] (Layered MMD) and TCA [53] (MMD). Firstly, in the measurement of distribution discrepancy, the above comparison model not only contains typical MMD and LMMD, but also has many improved versions. Secondly, the structure of the model is rich, which basically covers the commonly used feature extraction architecture, such as multi-branch CNN and AM. In particular, the network with physical knowledge is also included, which is helpful to explain the interpretability of the model. Meanwhile, in the following discussion and analysis, the diagnostic performance of the above methods is comprehensively and meticulously tested under various working conditions, which further ensures that the transfer diagnosis effect of the proposed methods is fully demonstrated.

4.2.1. Transfer fault diagnosis results under cross-load dataset

The single constant speed or load hardly conforms to the actual industrial scene. In order to measure the diagnostic effect of PISMEAN multi-variable cross-domain data, PU bearing data are selected to carry out multi-group transfer diagnostic tests. Meanwhile, set $N_t = 160$ or 120 to quantitatively analyze the sensitivity of the proposed method to the number of samples.

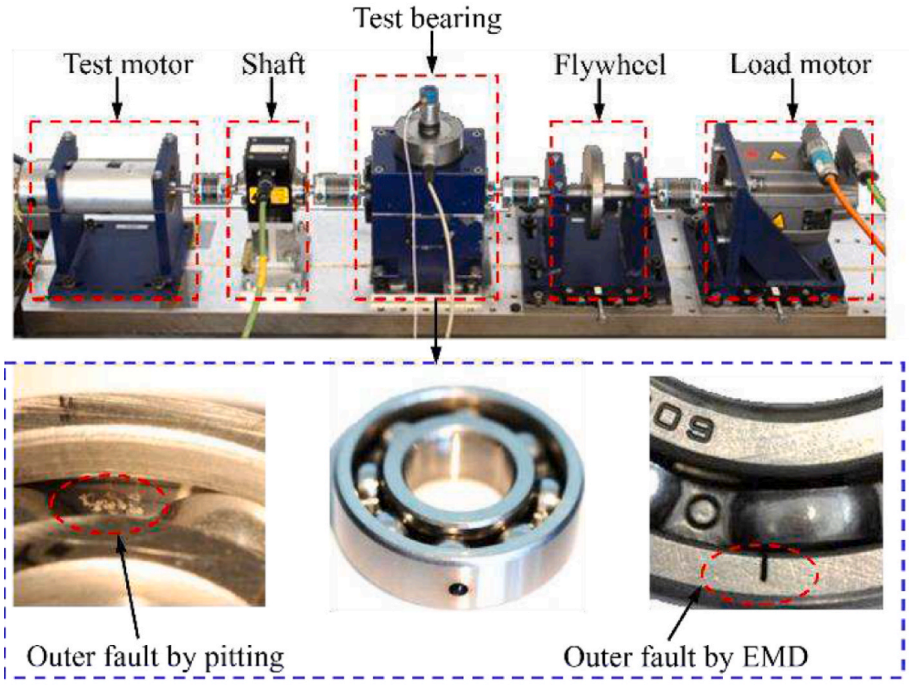


Fig. 6. PU simulation test bench.

Table 2
Description of PU datasets.

Transfer task	rotation speed	Load torque	Radial force	Description name	Fault category
A	1500	0.7	1000	N15_M07_F10	Health status-K001 (label 0), OR fault-KA01 (label 1), OR fault-KA06 (label 2), IR fault-KI08 (label 3), IR fault-KI17 (label 4).
B	900	0.7	1000	N09_M07_F10	
C	1500	0.1	1000	N15_M01_F10	
D	1500	0.7	400	N15_M07_F04	

As can be seen from Fig. 8, the accuracy of the proposed method maintains better stability overall in the four different diagnostic tasks, although it fluctuates with the change of working conditions. As shown in Fig. 8(a), when $N_t = 160$, the fault identification accuracy of PISMEAN is 85.62 %, 97.75 %, 96.88 % and 76.36 %, respectively. In addition to the high average accuracy, PISMEAN also shows relatively low standard deviations across different tasks (e.g., ±0.21, ±0.18, ±0.22, and ± 0.16), indicating stable diagnostic performance with minimal volatility. Compared with other methods, which exhibit larger fluctuations under domain shift, the proposed approach demonstrates stronger robustness and consistency across different working conditions. For example, in the B → C transfer task with the biggest data difference, compared with other IFD methods, the average diagnosis results of the proposed model are improved by 18.19 %, 4.84 %, 14.93 %, 3.74 %, 1.98 % and 5.55 %, respectively. Meanwhile, comparing the four groups of results and combining with Table 3, it can be seen that the fluctuation of rotating speed has a more significant impact on the distribution of data domain than the change of load. However, PISMEAN minimizes this effect, which also reflects that PIDConv can help the network adapt to the change of speed and extract fault features in different frequency bands by adaptive filtering characteristics.

Secondly, it can be seen from Fig. 8(b), limited by the original data, the generalization of data-driven IFD method has declined in all diagnostic tests. Encouragingly, the average result of PISMEAN is still above 85 %. Moreover, the standard deviation across multiple runs remains consistently low, indicating that the proposed method maintains stable performance even under limited data conditions. This can be attributed to the integration of physics-informed features and subdomain-level alignment mechanisms. Not surprisingly, the accuracy of TCA and MRANet is much lower than that of DDC, EWSNet, ISEANet and DSAN, which shows that when faced with complex data, the calculation method of local refinement can better capture the identifiable features of the domain and help improve the generalization performance of the model. It is worth noting that under the conditions of two training samples, the diagnosis effect of physical knowledge networks, such as ISEANet, EWSNet and PISMEAN, is better than that of networks that only rely on data, which also shows that the proposed method combined with physical knowledge and discrepancy calculation is helpful to enhance the generalization of networks.

In order to deeply analyze the ability of different approaches for specific fault categories, the confusion matrices of A → D are drawn respectively. As shown in Figs. 9 and 10. Firstly, by analyzing Fig. 9, it can be seen that almost all methods accurately identify the healthy bearing because the characteristic difference between the healthy bearing (label = 0) and the faulty bearing is more obvious. Secondly, networks with poor diagnostic performance, such as TCA and MRANet, confound the features of similar faults, indicating that MMD seems unable to accurately compute fine-grained local distinctions, despite being able to quantify distributional discrepancy from a global perspective. As shown in Fig. 9(g), PISMEAN can further learn the domain distribution state of high-dimensional space by relying on LMOMD, and the misjudgment rate is stably controlled at 2.5 %.

On the other hand, by analyzing Fig. 10, it can be seen that with the decrease of training samples, the methods that rely on data features, such as TCA, MRANet, DDC and DSAN, have significantly improved the misjudgment rate of different fault types (label = 1,2,3,4), which shows that IFD will seriously restrict its feature extraction ability under the condition of lack of data. However, PISMEAN enhances the physical information in the original signal by multi-mapping kernel filtering through PIDConv, and guided the network to preferentially extract the

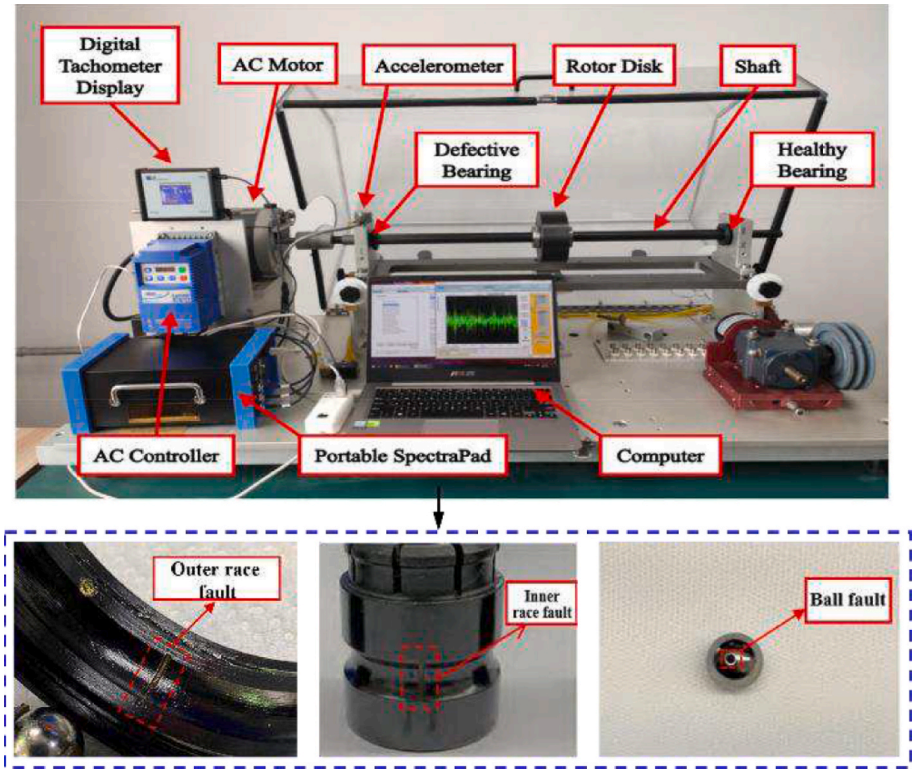


Fig. 7. MFS simulation test rig.

Table 3
Description of MFS datasets.

Motor end	Far end	Defect	Fault size	Rotational speed	label
$E_S/F_S/G_S$	$E_T/F_T/G_T$	OR	0.6	1200/1400/1600	0
$E_S/F_S/G_S$	$E_T/F_T/G_T$	OR	1.2	1200/1400/1600	1
$E_S/F_S/G_S$	$E_T/F_T/G_T$	OR	1.8	1200/1400/1600	2
$E_S/F_S/G_S$	$E_T/F_T/G_T$	IR	0.6	1200/1400/1600	3
$E_S/F_S/G_S$	$E_T/F_T/G_T$	IR	1.2	1200/1400/1600	4
$E_S/F_S/G_S$	$E_T/F_T/G_T$	IR	1.8	1200/1400/1600	5
$E_S/F_S/G_S$	$E_T/F_T/G_T$	Ball	1.2	1200/1400/1600	6
$E_S/F_S/G_S$	$E_T/F_T/G_T$	OR-IR	0.6–0.6	1200/1400/1600	7
$E_S/F_S/G_S$	$E_T/F_T/G_T$	OR-IR	0.6–1.2	1200/1400/1600	8

key features of different frequency bands representing bearing faults. Therefore, the proposed method can accurately identify all defect including different fault degrees, which indicates that PISMEAN has better transfer diagnosis stability.

4.2.2. *Transfer fault diagnosis results under MFS datasets*

With the operation of the equipment, the initial fault often induces and evolves into a compound fault with multiple defects. At this time, the collected vibration signals show greater distribution differences, and the internal structure of the data interferes with each other, which is difficult to describe. Meanwhile, the domain distribution of faulty bearings in different parts is also different, so it is more challenging for IFD to study mixed fault diagnosis. Accordingly, the MFS is used to collect single and compound data at different speeds, which correspond

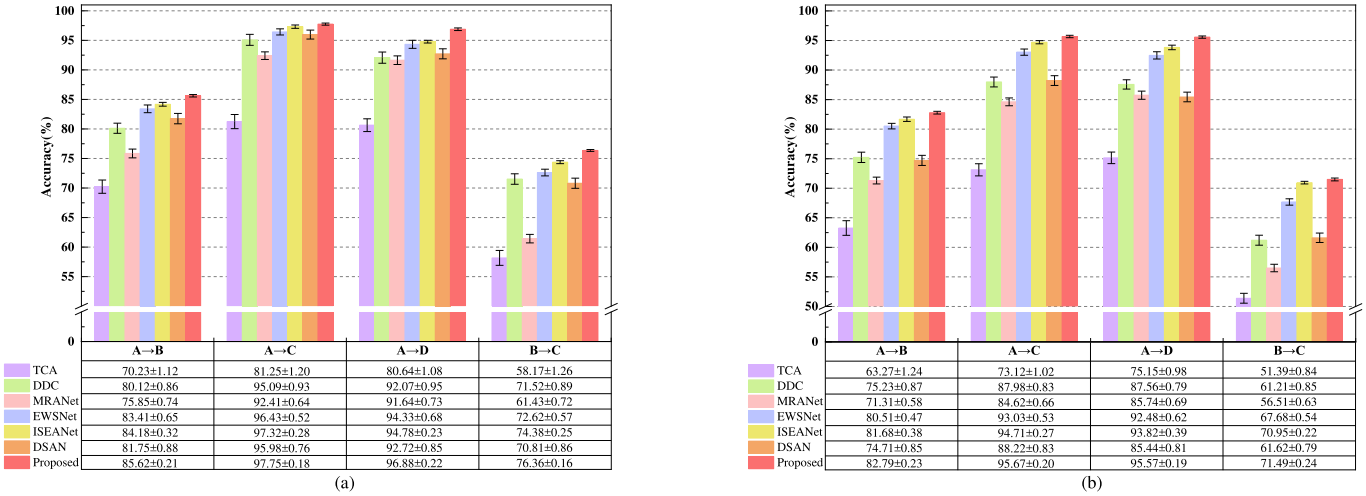


Fig. 8. Transfer fault diagnosis accuracy under different samples. (a) $N_t = 160$, (b) $N_t = 120$.

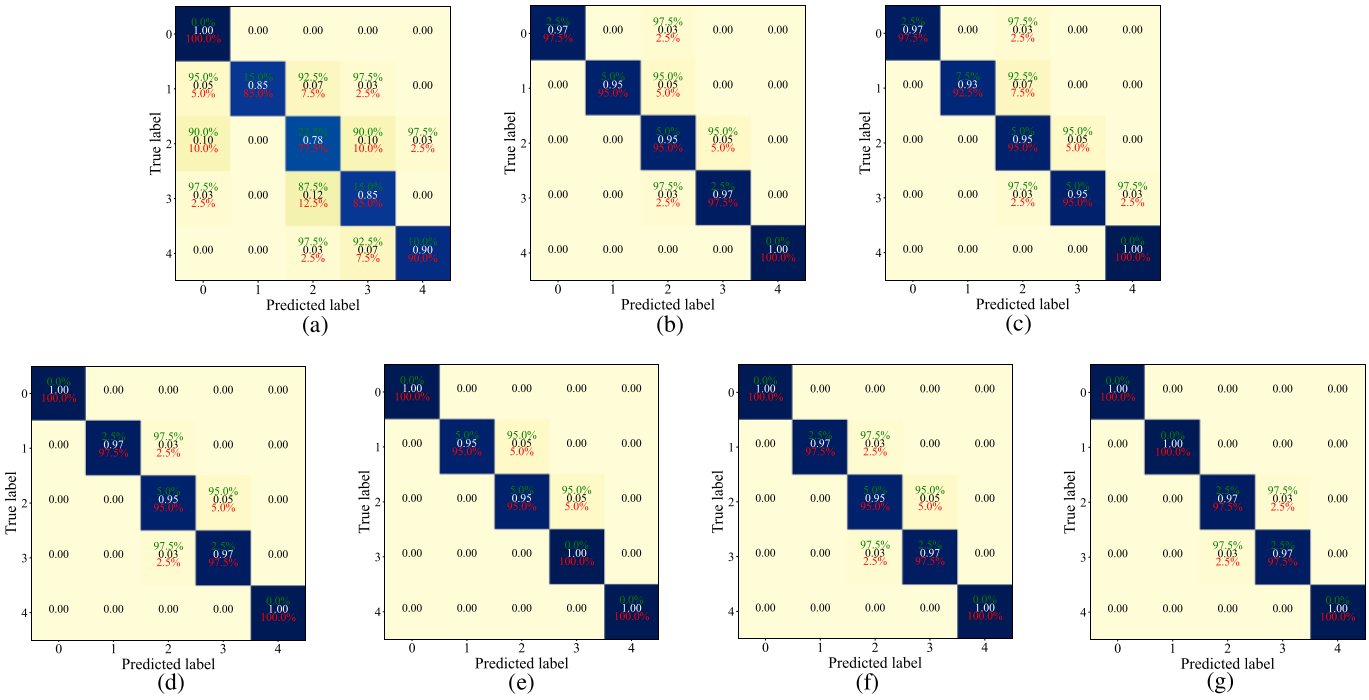


Fig. 9. Confusion matrix at $N_t = 160$. (a) TCA, (b) DDC, (c) MRANet, (d) EWSNet, (e) ISEANet, (f) DSAN (g) PISMEAN.

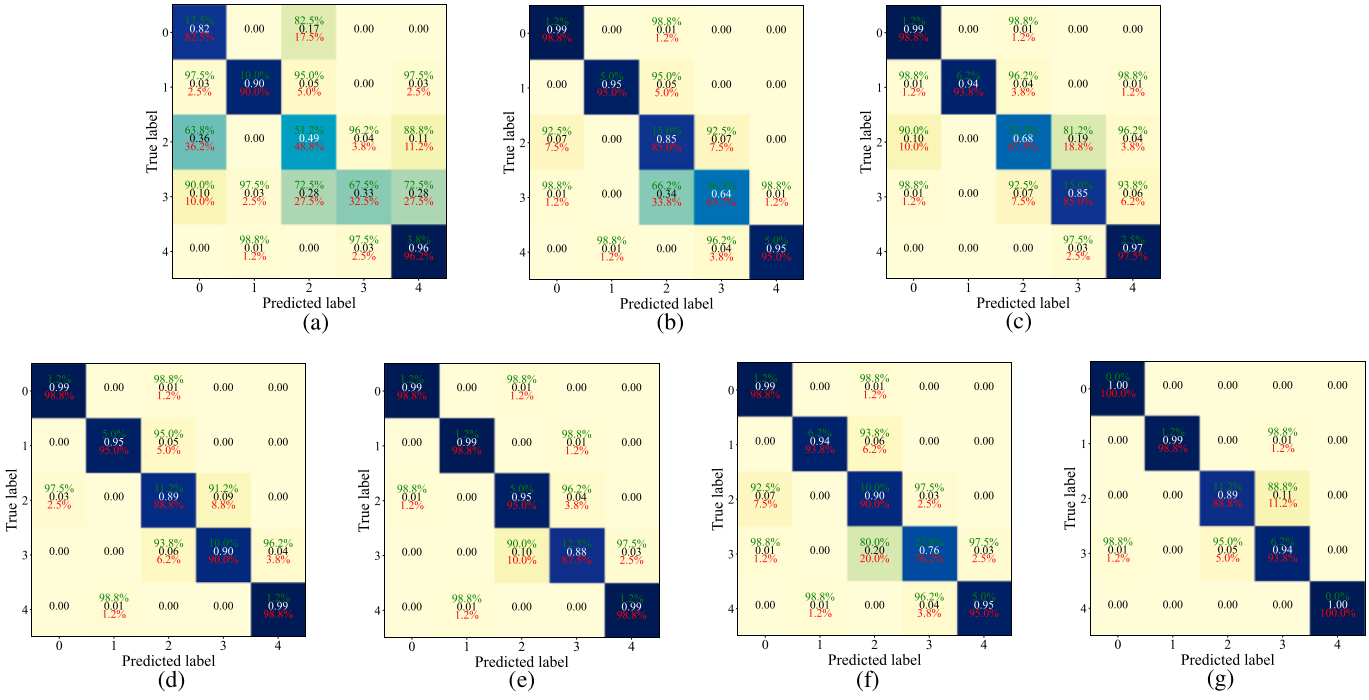


Fig. 10. Confusion matrix at $N_t = 120$. (a) TCA, (b) DDC, (c) MRANet, (d) EWSNet, (e) ISEANet, (f) DSAN (g) PISMEAN.

to the datasets of E_S, F_S, G_S, E_T, F_T, G_T.

Analyzing Table 4, it can be seen that PISMEAN shows better test results in six groups of diagnosis tasks, with an average accuracy of 84.13 %. Compared with the other six IFD methods, the recognition result is improved by 26.52 %, 12.96 %, 17.07 %, 6.72 %, 5.0 % and 11.76 % respectively. Obviously, the diagnostic accuracy of TCA and MRANet is much lower than that of DDC, EWSNet, ISEANet, DSAN and PISMEAN, which further indicates that in the face of cross-domain diagnosis with complex data, the generalization performance of the network can be enhanced by improving the discrepancy metric and embedding prior physical knowledge.

Secondly, although MFS datasets are more difficult, the average recognition accuracy of the proposed method remains at 80 %, which fully shows that LMOMD can accurately calculate the distribution discrepancy of multiple subdomains in high-dimensional space. Meanwhile, the average diagnostic accuracy of ISEANet and EWSNet is above 75 %, which also suggests that the physical knowledge network has potential application.

Table 4
Diagnostic accuracy on MFS dataset.

Methods	Experimental dataset type						
	$E_S \rightarrow E_T$	$F_S \rightarrow F_T$	$G_S \rightarrow G_T$	$E_S \rightarrow F_T$	$E_S \rightarrow G_T$	$F_S \rightarrow G_T$	Average
TCA	62.53	61.26	64.51	55.28	45.24	56.84	57.61 ± 6.38
DDC	76.47	75.82	77.15	68.27	61.49	67.84	71.17 ± 5.75
MRANet	70.21	73.48	74.63	62.31	56.33	65.42	67.06 ± 6.44
EWSNet	83.65	82.76	85.53	72.47	68.79	71.26	77.41 ± 6.71
ISEANet	84.68	85.83	83.26	75.44	71.32	74.24	79.13 ± 5.65
DSAN	78.15	76.42	79.61	67.96	62.75	69.34	72.37 ± 6.10
PISMEAN	88.71	91.65	89.48	78.54	76.63	79.81	84.13 ± 5.95

In order to analyze the DA performance of different methods more intuitively, $F_S \rightarrow F_T$ is selected and T-SNE algorithm is used to map features to low-dimensional space, which is convenient for visual analysis. On this basis, the kernel probability density curve is used to quantify the discrepancy calculation effect of MMD, LMMD, ILMMD, LMOMD. The result is shown in Fig. 11.

From Fig. 11(a), the original source and target domain data are randomly distributed in the feature space, and there is no regularity at all. The probability density of source domain (black solid line) and target domain (red dashed line) in Fig. 11(a') has a large domain deviation, which also explains that MFS mixed fault dataset is more difficult for IFD method.

Firstly, by analyzing Fig. 11(b)–(g), it can be seen that the identification effect of six comparison methods for single fault is obviously better than that of compound fault, which indicates that the coupling between faults further confuses the characteristic information and makes it difficult for the network to distinguish the fault. Secondly, by analyzing Fig. 11(b')–(g'), it is not difficult to find that the adaptive performance of local discrepancy metric is better than that of global domain metric. Therefore, for complex data, especially those containing homogeneous and compound fault signals, it is necessary to consider the distance between subdomains in a finer granularity, so as to improve the accuracy of model identification. Similarly, as can be seen from Fig. 11(h'), PISMEAN can further describe the internal structural relationship of data in high-dimensional space by improving LMMD, which significantly shortens the center distance of different domains. Finally, as shown in Fig. 11(h), the proposed method benefits from the fact that PIDConv empowers physical information to extract network features, and almost all fault types are correctly identified. Meanwhile, compared with the traditional first-order moment calculation method, LMOMD effectively improves the SA, reduces the intra-class distance and expands the inter-class decision boundary.

4.3. Ablation study

In order to further analyze the influence of different modules on PISMEAN, two different transfer tasks, $E_S \rightarrow E_T$ and $F_S \rightarrow F_T$, are used to conduct ablation experiments, and the training sample is $N_t = 170$. The specific comparison networks are as follows:

(1) M1: Removes PIDConv while retaining LMOMD, aiming to evaluate the impact of physical knowledge on feature extraction and model interpretability.
(2) M2: Retains PIDConv but replaces LMOMD with LMMD to assess the effect of different subdomain-level discrepancy metrics on model performance.
(3) M3: Retains PIDConv but replaces LMOMD with MMD to compare the influence of feature alignment strategies with different granularities.
(4) M4: Removes both the PIDConv and LMOMD. This model adopts standard convolution and traditional MMD for distribution adaptation, serving as a baseline model.

As can be seen from Fig. 12(a) and (b), the proposed modules in PISMEAN have clear and complementary contributions to transfer diagnosis performance. Starting from M4, which serves as the most basic model, the average classification accuracies are only 63.21 % and 60.12 % under the $E_S \rightarrow E_T$ and $F_S \rightarrow F_T$ tasks, respectively. This highlights the limited generalization capability of data-driven CNN under complex domain shifts. When LMOMD is introduced in M1, while keeping the conventional convolutional backbone, the accuracy improves by 5.93 % and 5.61 % respectively, indicating that high-order subdomain alignment can significantly enhance DA performance. Without the guidance of prior physical knowledge, the accuracy of the two kinds of transfer diagnosis decreased by 19.57 % and 25.92 % respectively. Further improvements are seen in M2, where PIDConv is preserved and LMOMD is replaced with LMMD. Compared to M3 (MMD), the use of subdomain-level alignment (LMMD) improves average accuracy by 3.02 %, suggesting that fine-grained alignment is more effective than global statistics in complex fault scenarios.

PISMEAN combines the advantages of the two to further enhance the generalization performance of the network, which fully demonstrates that this study is a useful exploration of the diagnosis method of physical-informed transfer. Specifically, it includes the following characteristics: (1) PIDConv effectively enhances multi-scale feature extraction by incorporating domain-specific prior knowledge through adaptive filtering and frequency-aware kernel initialization. (2) LMOMD can effectively mine the discrepancy of domain distribution in high-dimensional space, and further improve the generalization performance of PISMEAN compared with the first-order mean value.

4.4. Interpretable analysis of PISMEAN

As mentioned in Section 2, most of the traditional IFD methods used to explain the advanced features of network structure design from the classification accuracy or experimental results, but this subjective description seems to be unable to fully reveal the internal operation law of the network, resulting in a black-box in the process of feature extraction. In order to illustrate the advantages of the proposed method in detail, this section will analyze the dynamic adjustment effect of adaptive filtering kernel in PISMEAN from the perspectives of prior explainability of PIDConv and posterior visualization of convolution weight respectively to improve the interpretability of the diagnosis process. Accordingly, two groups of transfer diagnosis tasks A → C and $E_S \rightarrow E_T$ in MSF and PU datasets are selected as observation objects, and three bandpass filtering kernels in PIDConv are randomly selected, and their effects on the frequency response of vibration signals are plotted.

As shown in Fig. 13, the corresponding filtering kernels of different fault types (IR-OR, IR) are different in PIDConv, which indicates that the physical layer can adapt to the convolution mapping process, and adaptively adjust the filtering position according to the difference of physical information, showing multi-scale characteristics, breaking through the limitation of single bandpass filtering structure in traditional signal processing methods, and being more suitable for changeable cross-domain working conditions.

Firstly, as shown in Fig. 13(a)–(c), all three filter kernels show stable and symmetrical oscillation waveforms, indicating that they have excellent local properties in time domain, i.e., they actively adapt to the changes of different frequency bands of signals. At the same time, by analyzing Fig. 13(a')–(c'), it can be seen that the original signals are accurately extracted in high-energy concentrated frequency bands in different frequency ranges through dynamic filtering of PIDConv, and

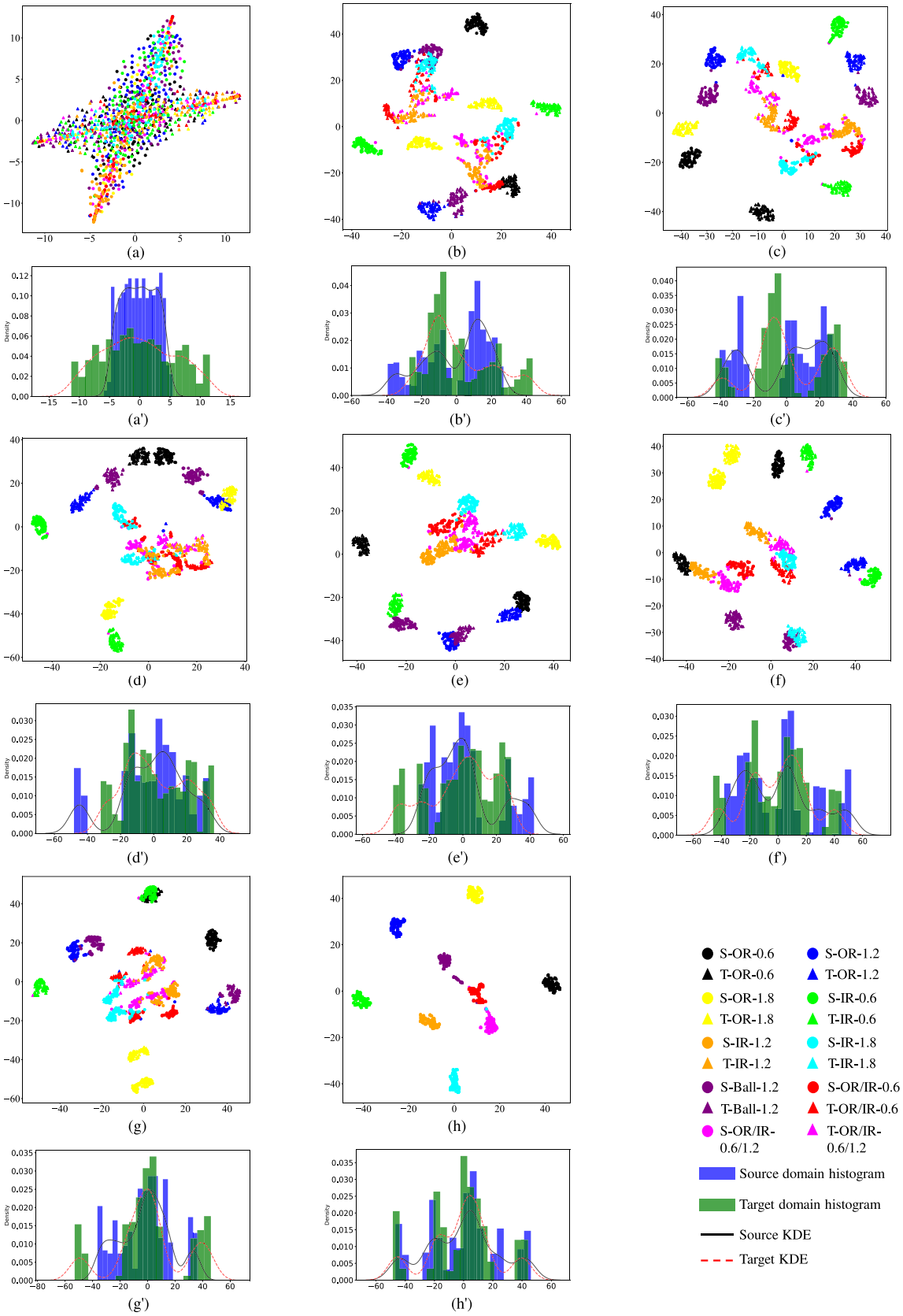


Fig. 11. Visualization of feature distribution. (a)→(h) are raw data, TCA, DDC, MRANet, EWSNet, ISEANet, DSAN, PISMEAN. (a')→(h') are the KDE and histograms corresponding to each method.

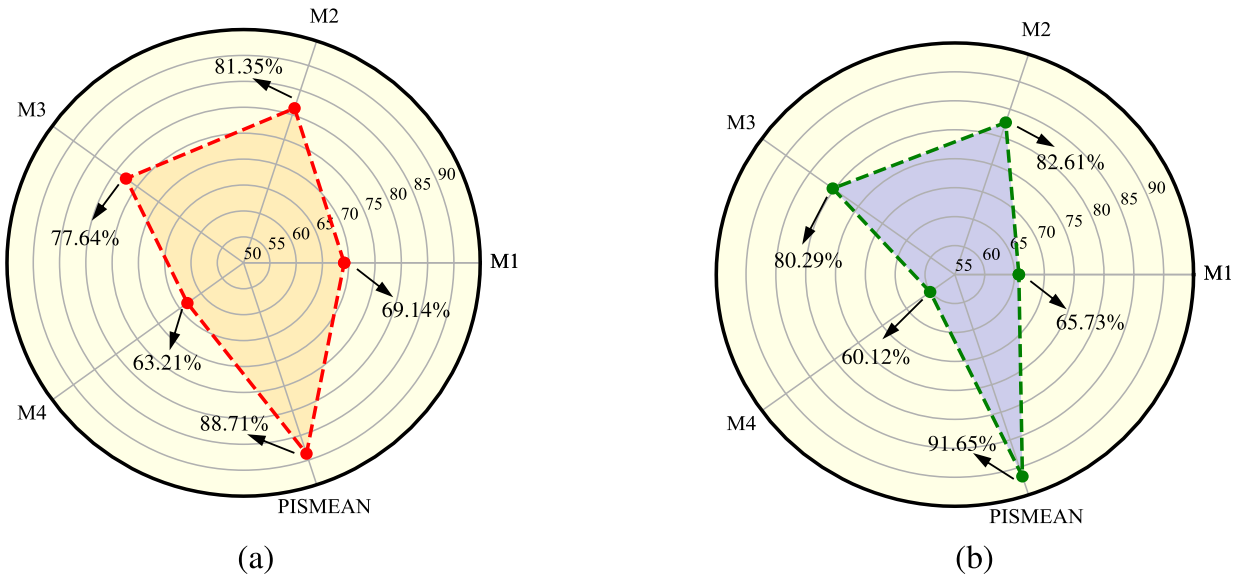


Fig. 12. Ablation experiment test results. (a) represent $E_S \rightarrow E_T$, (b) is $F_S \rightarrow F_T$.

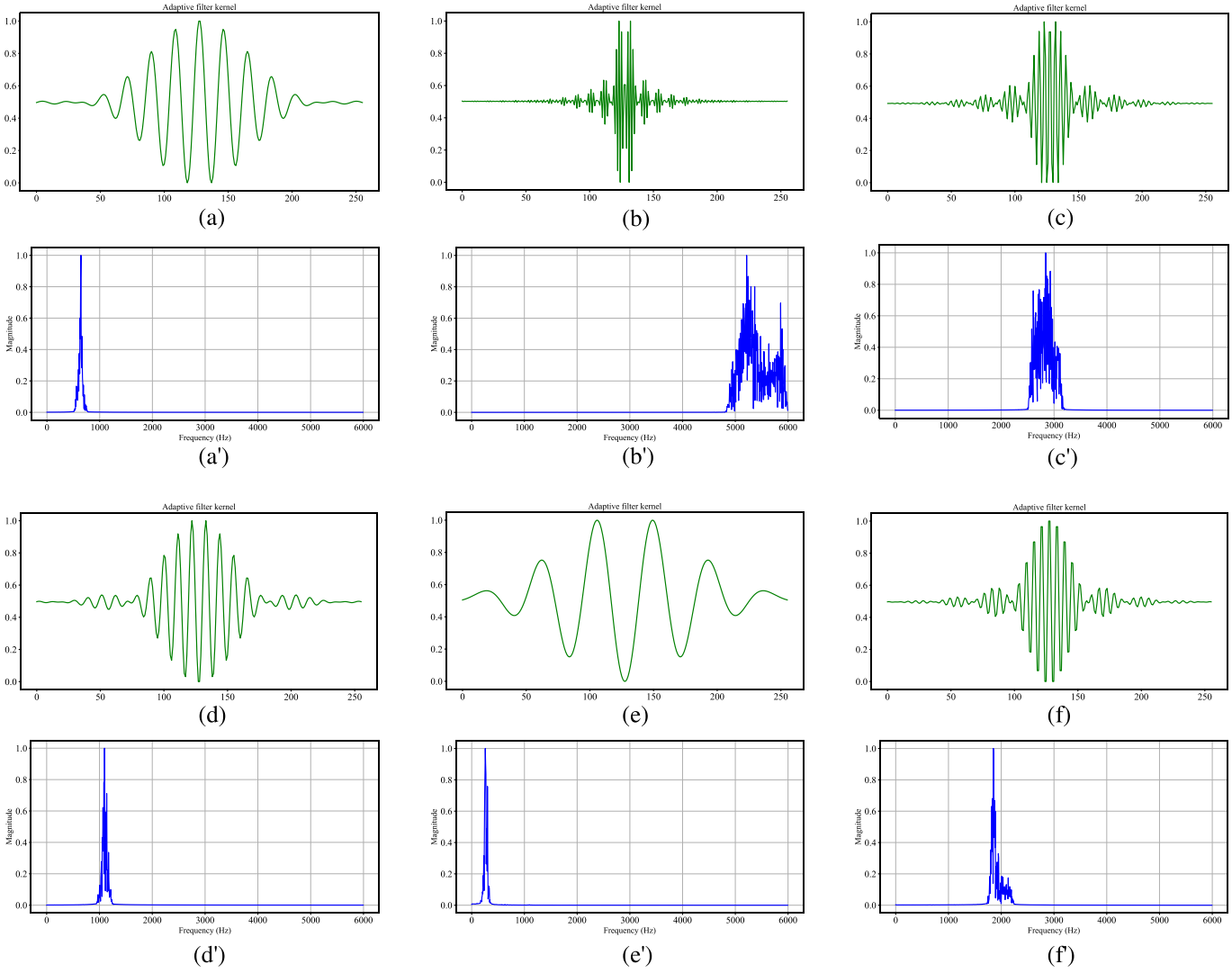


Fig. 13. Adaptive filtering kernel and signal frequency domain representation. (a)→(c) represent Compound fault, (d)→(f) is Single fault.

they do not interfere with each other, which also explains that PISMEAN can help the network extract key fault features by decoupling compound faults. Secondly, by comparing and analyzing Fig. 13 (d)–(f), the PID-Conv also shows high-efficiency perception for single fault signals. Compared with compound faults, the frequency band distribution of single fault signals is more concentrated, so the filter kernel shows a slow change and pays more attention to the middle and low frequency domain. In the same way, the key physical information frequency bands in Fig. 13 (d')–(f') are reserved in a targeted way, thus reducing the interference of noise. The above results also show that, distinguishing

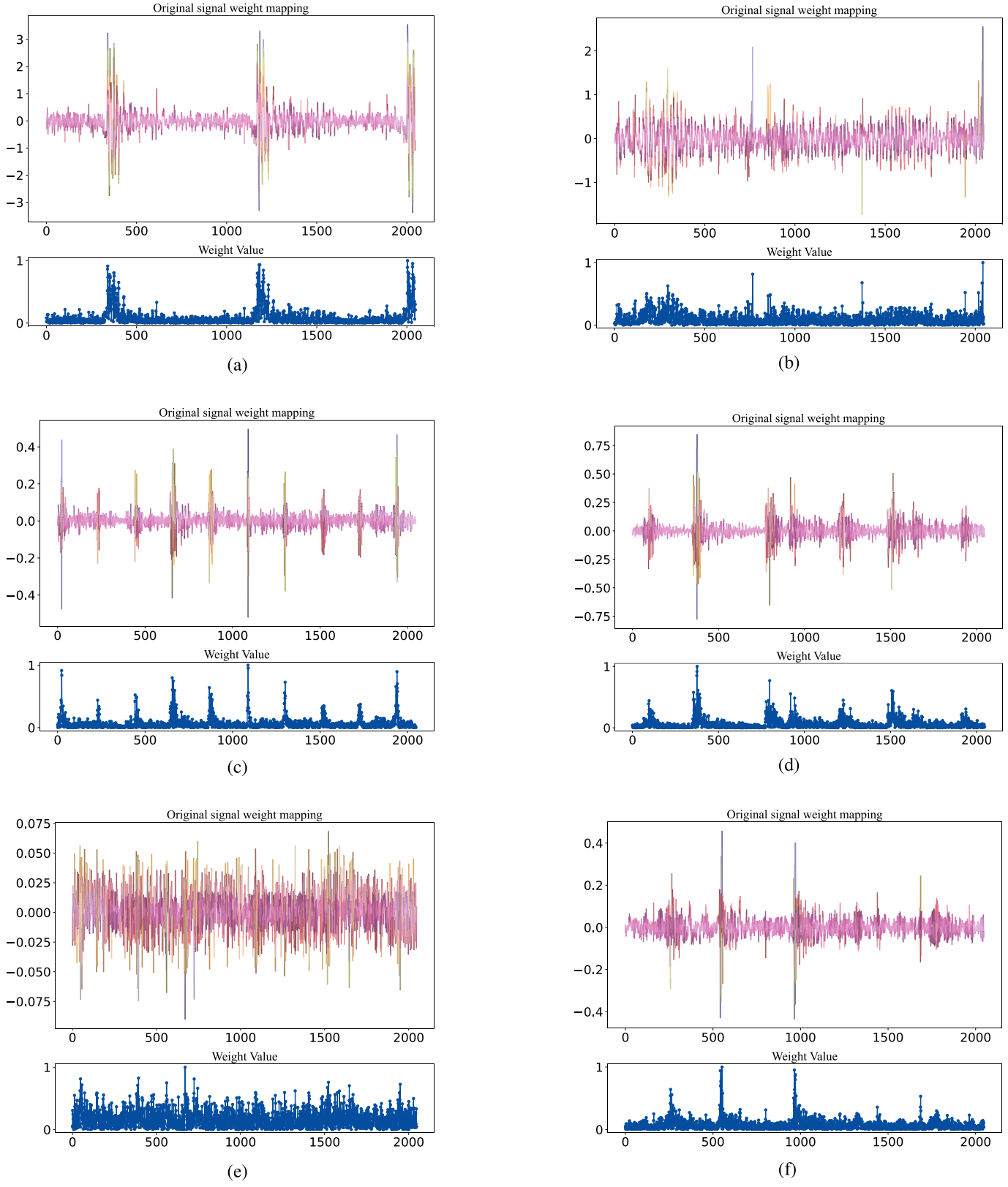


Fig. 14. Visualization of feature weight distribution. (a)→(b) represent OR, IR fault in PU datasets, (c)→(f) are OR, IR, Ball, IR-OR fault in MFS datasets.

from AM, PIDConv embeds prior physical knowledge into the network and realizes adaptive adjustment through BP.

On the other hand, the weight distribution results of key information in the subsequent convolution layer are drawn by CAM [54], which can be explained and analyzed from the posterior results. Similarly, different fault signals of PU datasets and MFS datasets are selected for visualization, and the results are shown in Fig. 14.

Thanks to the embedding and guidance of physical information by PIDConv, the subsequent convolution layer more accurately assigns high weights to the key impulse positions of vibration signals, which is beneficial to network learning fault features. It can be seen from Fig. 14 (a) and (b), the activation areas of the OR and IR are obviously different, which is because the excitation generated by the defect of the OR of the bearing has obvious periodicity, while the IR rotates with the shaft and is modulated by the speed, which affects the continuity of impact information. However, by analyzing the weight distribution, the proposed method still allocates weights adaptively and pays more attention to important features. Secondly, by comparing Fig. 14 (c)–(f), it can be seen that PISMEAN has carried out targeted data analysis and weight distribution, which shows that the network has the ability of fault decoupling through the guidance of physical information. Finally, it is worth noting that the original signal is very important for the effect of network feature extraction. For example, from Fig. 14 (e), the weight distribution of ball fault is almost irregular, which also shows that the knowledge contained in the initial signal is more helpful to improve the diagnosis effect of the network than the subsequent feature mapping. The above results indicate that PISMEAN is interpretable, and its feature learning process can actively extract key fault features according to the dynamic adjustment of PIDConv, which provides a more accurate feature for subsequent calculation of subdomain distribution discrepancy.

5. Conclusions

In order to improve the performance of the DA in complex working conditions, this paper proposes a new IFD method PISMEAN, which embeds physical knowledge into the network architecture and enhances feature alignment through multi-order moments. Firstly, the PIDConv based on physical information is introduced. This structure is driven by physical prior knowledge to realize adaptive dynamic bandpass filtering, which enhances the ability of the network to extract key features directly from the original vibration signal and enhances the interpretability of the model. Secondly, a new subdomain discrepancy metric, LMOMD, is proposed to align the spatial distribution between the source and target domain by considering higher-order statistical moment. LMOMD improves the sensitivity of the model to fine-grained domain transfer and enhances the generalization ability of PISMEAN under different working conditions. Finally, quantitative analysis of domain distribution discrepancy, ablation experiments and interpretable analysis are carried out. According to results, PISMEAN outperforms other IFD approaches in terms of transfer generalization.

To further improve the generalization and interpretability of fault diagnosis under complex conditions, it is necessary not only to enhance physical knowledge embedding in network design, but also to explore more effective integration of physical mechanisms and data-driven models. Therefore, the authors plan to conduct further research on physics-guided and adaptive domain alignment methods for intelligent fault diagnosis.

CRediT authorship contribution statement

Bin Liu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Changfeng Yan:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Investigation, Funding acquisition. **Chao He:** Writing – review & editing, Software, Methodology, Formal analysis. **Ming Lv:** Validation, Formal analysis, Data curation. **Jianxiong Wei:** Formal analysis, Data curation. **Lixiao Wu:** Writing – review & editing, Supervision, Resources, Project administration, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

[1] Y. Xiao, H. Shao, J. Wang, S. Yan, B. Liu, Bayesian variational transformer: a generalizable model for rotating machinery fault diagnosis, Mech. Syst. Sig. Process. 207 (2024) 110936.
[2] M. Cerrada, R.V. Sánchez, C. Li, F. Pacheco, D. Cabrera, J. Valente de Oliveira, R. E. Vásquez, A review on data-driven fault severity assessment in rolling bearings, Mech. Syst. Sig. Process. 99 (2018) 169–196.
[3] S. Li, Y. Peng, Y. Shen, S. Zhao, H. Shao, G. Bin, Y. Guo, X. Yang, C. Fan, Rolling bearing fault diagnosis under data imbalance and variable speed based on adaptive clustering weighted oversampling, Rel. Eng. Syst. Saf. 244 (2024) 109938.
[4] J. Wang, J. Zheng, H. Pan, J. Tong, Q. Liu, Refined composite multiscale slope entropy and its application in rolling bearing fault diagnosis, ISA Trans. 152 (2024) 371–384.
[5] S. Wang, G. Lian, C. Cheng, H. Chen, A novel method of rolling bearings fault diagnosis based on singular spectrum decomposition and optimized stochastic configuration network, Neurocomputing 574 (2024) 127278.
[6] M. Khan, W. Gueaieb, A. El Saddik, S. Kwon, MSER: multimodal speech emotion recognition using cross-attention with deep fusion, Expert Syst. Appl. 245 (2024) 122946.
[7] P. Oza, V.A. Sindagi, V. Vs, V.M. Patel, Unsupervised domain adaptation of object detectors: a survey, ITPAM. 46 (2024) 4018–4040.
[8] H. Bilal, M.S. Obaidat, M. Shamrooz Aslam, J. Zhang, B. Yin, K. Mahmood, Online fault diagnosis of industrial robot Using IoRT and hybrid deep learning techniques: an experimental approach, IEEE INTERNET THINGS. 11 (2024) 31422–31437.
[9] Z. Dong, D. Zhao, L. Cui, An intelligent bearing fault diagnosis framework: one-dimensional improved self-attention-enhanced CNN and empirical wavelet transform, Nonlinear Dyn. 112 (2024) 6439–6459.
[10] Y. Xu, K. Feng, X. Yan, X. Sheng, B. Sun, Z. Liu, R. Yan, Cross-Modal fusion convolutional neural networks with online soft-label training strategy for mechanical fault diagnosis, IEEE Trans. Ind. Inf. 20 (2024) 73–84.
[11] Y. An, K. Zhang, Y. Chai, Z. Zhu, Q. Liu, Gaussian mixture variational-based transformer domain adaptation fault diagnosis method and its application in bearing fault diagnosis, IEEE Trans. Ind. Inf. 20 (2024) 615–625.
[12] Z. Zhu, Y. Lei, G. Qi, Y. Chai, N. Mazur, Y. An, X. Huang, A review of the application of deep learning in intelligent fault diagnosis of rotating machinery, Measurement 206 (2023) 112346.
[13] Y. Peng, H. Shao, Y. Xiao, S. Yan, J. Wang, B. Liu, A systematic review on interpretability research of intelligent fault diagnosis models, Meas. Sci. Technol. 36 (2024) 012009.
[14] C. He, H. Shi, X. Liu, J. Li, Interpretable physics-informed domain adaptation paradigm for cross-machine transfer diagnosis, Knowl. Based Syst. 288 (2024) 111499.
[15] Y. Qin, H. Liu, Y. Mao, Faulty rolling bearing digital twin model and its application in fault diagnosis with imbalanced samples, Adv. Eng. Informatics. 61 (2024) 102513.
[16] H. Ahmad, W. Cheng, J. Xing, W. Wang, S. Du, L. Li, R. Zhang, X. Chen, J. Lu, Deep learning-based fault diagnosis of planetary gearbox: a systematic review, J. Manuf. Syst. 77 (2024) 730–745.
[17] Z. Fan, Q. Xu, C. Jiang, S.X. Ding, Deep mixed domain generalization network for intelligent fault diagnosis under unseen conditions, IEEE Trans. Ind. Electron. 71 (2024) 965–974.
[18] T. Gao, J. Yang, Q. Tang, A multi-source domain information fusion network for rotating machinery fault diagnosis under variable operating conditions, Inf. Fusion 106 (2024) 102278.
[19] J. Meng, C. Yan, Z. Wang, T. Wen, G. Chen, L. Wu, Health condition identification of rolling element bearing based on gradient of features matrix and MDDCs-MRSVD, IEEE Trans. Instrum. Meas. 71 (2022) 1–13.
[20] Q. Qian, Y. Wang, T. Zhang, Y. Qin, Maximum mean square discrepancy: a new discrepancy representation metric for mechanical fault transfer diagnosis, Knowl. Based Syst. 276 (2023) 110748.

[21] Y. Chen, C. Yang, Y. Zhang, Y. Li, Deep conditional adaptation networks and label correlation transfer for unsupervised domain adaptation, Pattern Recogn. 98 (2020) 107072.
[22] Q. Qian, J. Luo, Y. Qin, Adaptive intermediate class-wise distribution alignment: a universal domain adaptation and generalization method for machine fault diagnosis, IEEE Trans. Neural Netw. Learn. Syst. (2024) 1–15.
[23] L. Cui, Y. Xiao, D. Liu, H. Han, Digital twin-driven graph domain adaptation neural network for remaining useful life prediction of rolling bearing, Rel. Eng. Syst. Saf. 245 (2024) 109991.
[24] B. Yang, Y. Lei, X. Li, C. Roberts, Deep targeted transfer learning along designable adaptation trajectory for fault diagnosis across different machines, IEEE Trans. Ind. Electron. 70 (2023) 9463–9473.
[25] Q. Qian, J. Zhou, Y. Qin, Relationship transfer domain generalization network for rotating machinery fault diagnosis under different working conditions, IEEE Trans. Ind. Inf. 19 (2023) 9898–9908.
[26] X. Yu, Y. Wang, Z. Liang, H. Shao, K. Yu, W. Yu, An adaptive domain adaptation method for rolling bearings' fault diagnosis fusing deep convolution and self-attention networks, IEEE Trans. Instrum. Meas. 72 (2023) 1–14.
[27] X. Chen, H. Shao, Y. Xiao, S. Yan, B. Cai, B. Liu, Collaborative fault diagnosis of rotating machinery via dual adversarial guided unsupervised multi-domain adaptation network, Mech. Syst. Sig. Process. 198 (2023) 110427.
[28] Q. Li, L. Chen, L. Kong, D. Wang, M. Xia, C. Shen, Cross-domain augmentation diagnosis: an adversarial domain-augmented generalization method for fault diagnosis under unseen working conditions, Rel. Eng. Syst. Saf. 234 (2023) 109171.
[29] Y. Deng, J. Lv, D. Huang, S. Du, Combining the theoretical bound and deep adversarial network for machinery open-set diagnosis transfer, Neurocomputing 548 (2023) 126391.
[30] J. Wang, Z. Zhang, Z. Liu, B. Han, H. Bao, S. Ji, Digital twin aided adversarial transfer learning method for domain adaptation fault diagnosis, Rel. Eng. Syst. Saf. 234 (2023) 109152.
[31] Y. Xiao, H. Shao, Z. Min, H. Cao, X. Chen, J. Lin, Multiscale dilated convolutional subdomain adaptation network with attention for unsupervised fault diagnosis of rotating machinery cross operating conditions, Measurement 204 (2022) 112146.
[32] J. Liu, L. Wan, F. Xie, Y. Sun, X. Wang, D. Li, S. Wu, Cross-machine deep subdomain adaptation network for wind turbines fault diagnosis, Mech. Syst. Sig. Process. 210 (2024) 111151.
[33] H. Shi, S. Cao, H. Zuo, J. Ma, C. Lin, Deep subdomain adversarial network with self-supervised learning for aero-engine high speed bearing fault diagnosis with unknown working conditions, Measurement 241 (2025) 115668.
[34] J. Zhuang, Y. Chen, X. Zhao, M. Jia, K. Feng, A graph-embedded subdomain adaptation approach for remaining useful life prediction of industrial IoT systems, IEEE Internet Things J. 11 (2024) 22903–22914.
[35] C. Luo, J. Wang, E. Zio, Q. Miao, Subdomain adaptation order network for fault diagnosis of brushless DC motors, IEEE Trans. Instrum. Meas. 73 (2024) 1–10.
[36] X. Yuan, D. Shi, N. Shi, Y. Li, P. Liang, L. Zhang, Z. Zheng, Intelligent fault diagnosis of rolling bearing based on an active federated local subdomain adaptation method, Adv. Eng. Informatics. 62 (2024) 102807.
[37] J. Fang, K. Chen, C. Liu, J. He, An explainable and robust method for fault classification and location on transmission lines, IEEE Trans. Ind. Inf. 19 (2023) 10182–10191.
[38] M.S. Kim, J.P. Yun, P. Park, Deep learning-based explainable fault diagnosis model with an individually grouped 1-D convolution for three-axis vibration signals, IEEE Trans. Ind. Inf. 18 (2022) 8807–8817.
[39] B. Chen, T. Liu, C. He, Z. Liu, L. Zhang, Fault diagnosis for limited annotation signals and strong noise based on interpretable attention mechanism, IEEE Sensors J. 22 (2022) 11865–11880.
[40] D. Yang, H.R. Karimi, L. Gelman, An explainable intelligence fault diagnosis framework for rotating machinery, Neurocomputing 541 (2023) 126257.
[41] M. Gwak, M.S. Kim, J.P. Yun, P. Park, Robust and explainable fault diagnosis with power-perturbation-based decision boundary analysis of deep learning models, IEEE Trans. Ind. Inf. 19 (2023) 6982–6992.
[42] B. Liu, C. Yan, Y. Liu, M. Lv, Y. Huang, L. Wu, ISEANet: an interpretable subdomain enhanced adaptive network for unsupervised cross-domain fault diagnosis of rolling bearing, Adv. Eng. Informatics. 62 (2024) 102610.
[43] C. He, H. Shi, R. Li, J. Li, Z. Yu, Interpretable modulated differentiable STFT and physics-informed balanced spectrum metric for freight train wheelset bearing cross-machine transfer fault diagnosis under speed fluctuations, Adv. Eng. Informatics. 62 (2024) 102568.
[44] S. Li, T. Li, C. Sun, X. Chen, R. Yan, WPConvNet: an interpretable wavelet packet kernel-constrained convolutional network for noise-robust fault diagnosis, IEEE Trans. Neural Netw. Learn. Syst. 35 (2024) 14974–14988.
[45] B. Liu, C. Yan, Y. Liu, Z. Wang, Y. Huang, L. Wu, Multiscale residual antinoise network via interpretable dynamic recalibration mechanism for rolling bearing fault diagnosis with few samples, IEEE Sensors J. 23 (2023) 31425–31439.
[46] Q. Li, H. Li, W. Hu, S. Sun, Z. Qin, F. Chu, Transparent operator network: a fully interpretable network incorporating learnable wavelet operator for intelligent fault diagnosis, IEEE Trans. Ind. Inf. 20 (2024) 8628–8638.
[47] J. Ou, Y. Li, C. Shen, Unlabeled PCA-shuffling initialization for convolutional neural networks, Appl. Intell. 48 (2018) 4565–4576.
[48] C. He, H. Shi, J. Si, J. Li, Physics-informed interpretable wavelet weight initialization and balanced dynamic adaptive threshold for intelligent fault diagnosis of rolling bearings, J. Manuf. Syst. 70 (2023) 579–592.
[49] J. Zhong, Y. Zheng, C. Ruan, L. Chen, X. Bao, L. Lyu, M-IPISincNet: an explainable multi-source physics-informed neural network based on improved SincNet for rolling bearings fault diagnosis, Inf. Fusion 115 (2025) 102761.
[50] J. Chen, W. Hu, D. Cao, Z. Zhang, Z. Chen, F. Blaabjerg, A meta-learning method for electric machine bearing fault diagnosis under varying working conditions with limited data, IEEE Trans. Ind. Inf. 19 (2023) 2552–2564.
[51] J. Luo, H. Shao, H. Cao, X. Chen, B. Cai, B. Liu, Modified DSAN for unsupervised cross-domain fault diagnosis of bearing under speed fluctuation, J. Manuf. Syst. 65 (2022) 180–191.
[52] S. Schwendemann, Z. Amjad, A. Sikora, Bearing fault diagnosis with intermediate domain based Layered Maximum Mean Discrepancy: a new transfer learning approach, Eng. Appl. Artif. Intell. 105 (2021) 104415.
[53] R. van de Sand, S. Corasaniti, J. Reiff-Stephan, Data-driven fault diagnosis for heterogeneous chillers using domain adaptation techniques, Control Eng. Pract. 112 (2021) 104815.
[54] X. Zhang, C. He, Y. Lu, B. Chen, L. Zhu, L. Zhang, Fault diagnosis for small samples based on attention mechanism, Measurement 187 (2022) 110242.